

Convex optimalization

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This document contains lecture notes from the course *Convex optimizations* in the winter semester 2025/26, taught by Michael Kompatscher. The outline and content of the notes correspond directly to the content of the lectures. The main sources for the course were book and slides by Stephen Boyd and Lieven Vandenberghe.

This document was created with the help of Gemini and may contain mistakes.

1 Introduction

1.1 Optimization problem

A *mathematical optimization problem*, or just *optimization problem*, has the form

$$\begin{aligned} & \text{minimize} && f_0(x) \\ & \text{subject to} && f_i(x) \leq b_i, \quad i = 1, \dots, m. \end{aligned}$$

Here the vector $x = (x_1, \dots, x_n)$ is the *optimization variable* of the problem, the function $f_0 : \mathbb{R}^n \rightarrow \mathbb{R}$ is the *objective function*, the functions $f_i : \mathbb{R}^n \rightarrow \mathbb{R}$, $i = 1, \dots, m$, are the (inequality) *constraint functions*, and the constants $b_1, \dots, b_m$ are the *limits*, or *bounds*, for the constraints. A vector x^* is called *optimal*, or a solution of the problem, if it has the smallest objective value among all vectors that satisfy the constraints: for any z with $f_1(z) \leq b_1, \dots, f_m(z) \leq b_m$, we have $f_0(z) \geq f_0(x^*)$.

1.2 Least square

A least-squares problem is an optimization problem with *no constraints* (i.e., $m = 0$) and an *objective which is a sum of squares of terms* of the form $a_i^T x - b_i$:

$$\text{minimize} \quad f_0(x) = \|Ax - b\|_2^2 = \sum_{i=1}^k (a_i^T x - b_i)^2.$$

Here $A \in \mathbb{R}^{k \times n}$ (with $k \geq n$), a_i^T are the rows of A , and the vector $x \in \mathbb{R}^n$ is the optimization variable.

The solution of a least-squares problem can be reduced to solving a set of linear equations,

$$(A^T A)x = A^T b,$$

so we have the analytical solution $x = (A^T A)^{-1} A^T b$. For least-squares problems we have good algorithms (and software implementations) for solving the problem to high accuracy, with very high reliability.

1.3 Linear programming

Another important class of optimization problems is *linear programming*, in which the *objective* and *all constraint* functions are *linear*:

$$\begin{aligned} & \text{minimize} && c^T x \\ & \text{subject to} && a_i^T x \leq b_i, \quad i = 1, \dots, m. \end{aligned} \tag{1.5}$$

Here the vectors $c, a_1, \dots, a_m \in \mathbb{R}^n$ and scalars $b_1, \dots, b_m \in \mathbb{R}$ are problem parameters that specify the objective and constraint functions.

1.4 Convex optimization

A *convex optimization* problem is one of the form:

$$\begin{aligned} & \text{minimize} && f_0(x) \\ & \text{subject to} && f_i(x) \leq 0, \quad i = 1, \dots, m \\ & && Ax = b \end{aligned}$$

where *variable* $x \in \mathbb{R}^n$, *equality constraints* are *linear*, functions $f_0, \dots, f_m$ are *convex*, i.e. satisfy for $\theta \in [0, 1]$

$$f_i(\theta x + (1 - \theta)y) \leq \theta f_i(x) + (1 - \theta)f_i(y)$$

The least-squares problem and linear programming problem are both *special cases* of the general convex optimization problem.

2 Convex sets

Notation. Let

- $x = (x_1, \dots, x_n) \in \mathbb{R}^n$
- $x \succeq y \iff \forall i = 1, \dots, n : x_i \geq y_i$
- $x \succ y \iff \forall i = 1, \dots, n : x_i > y_i$

Remark. Some $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ can be partial. (For example: $\log : \mathbb{R} \rightarrow \mathbb{R}$.)

Definition (Line segment). Let $x_1 \neq x_2 \in \mathbb{R}^n$ be two points.

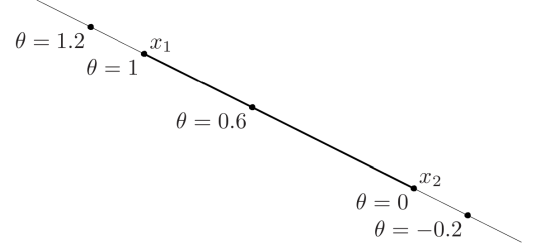
- The line through x_1 and x_2 consists of all points:

$$y = \theta x_1 + (1 - \theta)x_2, \quad \theta \in \mathbb{R}$$

- The line segment between x_1 and x_2 is obtained when:

$$y = \theta x_1 + (1 - \theta)x_2, \quad \theta \in [0, 1]$$

- Alternative form: $y = x_2 + \theta(x_1 - x_2)$, where x_2 is the base point and $x_1 - x_2$ is the direction vector.

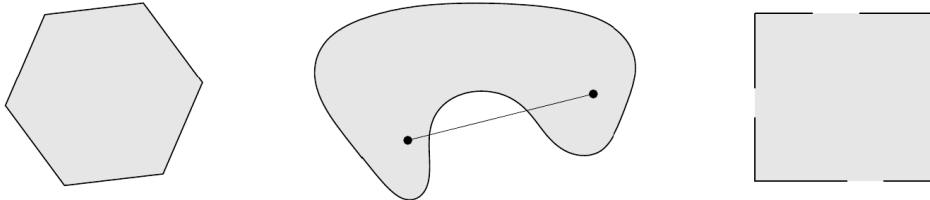


The line through x_1 and x_2 is described parametrically by $\theta x_1 + (1 - \theta)x_2$, where $\theta \in \mathbb{R}$. The line segment between x_1 and x_2 , which corresponds to $\theta \in [0, 1]$, is shown darker.

Definition (Affine set). A set $A \subseteq \mathbb{R}^n$ is affine if the line through any two distinct points in A lies in A , i.e. if $\forall x, y \in A$ and $\forall \theta \in \mathbb{R}$ we have $\theta x + (1 - \theta)y \in A$.

Definition (Convex set). A set $C \subseteq \mathbb{R}^n$ is convex if $\forall x, y \in C$ and for all $0 \leq \theta \leq 1$:

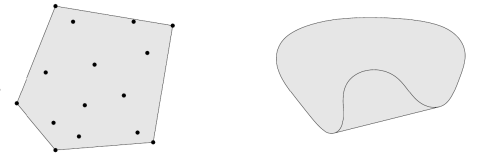
$$\theta x + (1 - \theta)y \in C.$$



Convex set (left) and two non-convex sets (middle, right).

Definition (Convex combination). We call a point of the form $\sum_{i=1}^k \theta_i x_i$, where $\theta_1 + \dots + \theta_k = 1$ and $\theta_i \geq 0, i = 1, \dots, k$, a convex combination of the points $x_1, \dots, x_k$.

Observation. A set is convex $\iff$ it contains every convex combination of its points.



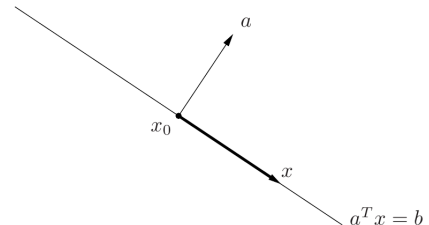
The convex hulls of two sets in $\mathbb{R}^2$.

Definition (Convex hull). The convex hull of a set C , denoted $\text{conv}(C)$, is the set of all convex combinations of points in C :

$$\text{conv}(C) = \left\{ \sum_{i=1}^k \theta_i x_i \in C, \theta_i \geq 0, i = 1, \dots, k, \theta_1 + \dots + \theta_k = 1 \right\}.$$

Definition (Hyperplane). Hyperplane is set of the form

$$\{x \mid a^T x = b\}, \text{ with } a \neq 0.$$



Definition (Halfspace). halfspace is set of the form

$$\{x \mid a^T x \leq b\}, \text{ with } a \neq 0.$$

Where a is the normal vector.

Remark. Hyperplanes are affine and convex; halfspaces are convex.

Observation. Halfspace is a convex set.

Proof. By definition. Let $C = \{x \mid a^T x \leq b\}$ and $x = \theta x_1 + (1 - \theta)x_2$, where $0 \leq \theta \leq 1$ and $x_1, x_2 \in C$. Now we can take:

$$a^T x = a^T(\theta x_1 + (1 - \theta)x_2) = a^T x_1 \theta + a^T x_2(1 - \theta).$$

Because $x_1, x_2 \in C$ it is true that $a^T x_1 \leq b$ and $a^T x_2 \leq b$, so:

$$a^T x_1 \theta + a^T x_2(1 - \theta) \leq b\theta + b(1 - \theta).$$

□

Definition (Balls). Ball with center x_c and radius r :

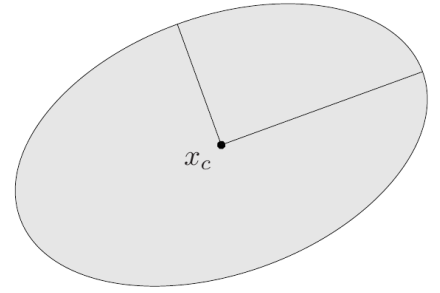
$$B(x_c, r) = \{x \mid \|x - x_c\|_2 \leq r\} = \{x_c + ru \mid \|u\|_2 \leq 1\}$$

Definition (Ellipsoid). Ellipsoid is set of the form

$$\{x \mid (x - x_c)^T P^{-1}(x - x_c) \leq 1\}$$

with $P \in S_{++}^n$ (i.e., P symmetric positive definite).

Another representation: $\{x_c + Au \mid \|u\|_2 \leq 1\}$ with A square and nonsingular.



An ellipsoid in $\mathbb{R}^2$ with the center x_c .

Definition (Norm). Norm is a function $\|\cdot\|$ that satisfies

- $\|x\| \geq 0$; $\|x\| = 0 \iff x = 0$,
- $\|tx\| = |t| \|x\|$ for $t \in \mathbb{R}$,
- $\|x + y\| \leq \|x\| + \|y\|$.

Definition (Norm cones). Norm cone is defined as

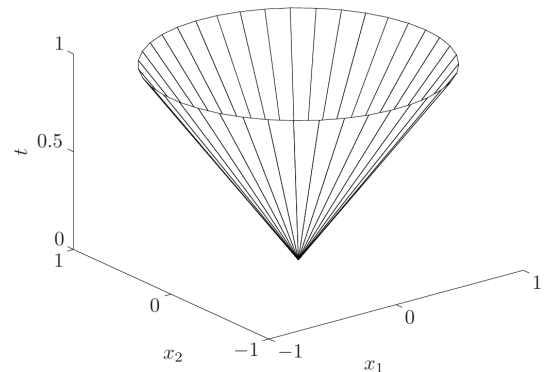
$$C = \{(x, t) \mid \|x\| \leq t\} \text{ and is convex.}$$

Observation. $C \subseteq \mathbb{R}^n$ is convex cone if

$$(\forall x, y) (\forall \theta_1, \theta_2 \geq 0) : \theta_1 x + \theta_2 y \in C.$$

Definition (Norm balls). Norm ball with center x_c and radius r :

$$B = \{x \mid \|x - x_c\| \leq r\}.$$



Boundary of second-order cone in $\mathbb{R}^3$.

Observation. *The norm ball is convex.*

Proof. For $x, y \in B$ and $0 \leq \theta \leq 1$:

$$\begin{aligned} \|\theta x + (1 - \theta)y - x_c\| &= \|\theta(x - x_c) + (1 - \theta)(y - x_c)\| = \\ \|\theta(x - x_c)\| + \|(1 - \theta)(y - x_c)\| &= \underbrace{\theta\|x - x_c\|}_{\leq r} + \underbrace{(1 - \theta)\|y - x_c\|}_{\leq r} \leq r \\ \implies \theta x + (1 - \theta)y &\in B. \end{aligned}$$

□

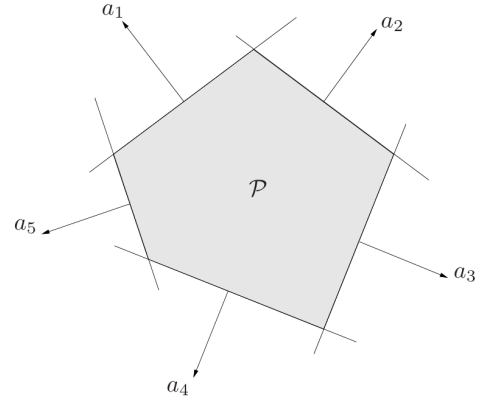
Definition (Convex polyhedra). *Polyhedron is solution set of finitely many linear inequalities and equalities*

$$S = \{x \mid Ax \preceq b, Cx = d\}$$

Where $A = \begin{pmatrix} a_1^T \\ \vdots \\ a_m^T \end{pmatrix} \in \mathbb{R}^{m \times n}$, $b = \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix}$, $C \in \mathbb{R}^{p \times n}$ and the

inequality sign ' $\preceq$ ' is componentwise.

It is an intersection of finite number of halfspaces and hyperplanes.



Polyhedron as the intersection of five halfspaces.

2.1 Checking if set C is convex:

1. Apply **definition**: show that $x_1, x_2 \in C, 0 \leq \theta \leq 1 \implies \theta x_1 + (1 - \theta)x_2 \in C$.
It is recommended only for very simple sets.
2. Use **convex functions**.
3. **Show that C is obtained from simple convex sets** by operations that preserve convexity:
 - (a) intersection,
 - (b) affine mapping,
 - (c) perspective function, ...

2.1.1 Intersection.

For $(C_i)_{i \in I}$ $C_i \subseteq \mathbb{R}^n$ is convex $\forall i \in I \implies C = \bigcap_{i \in I} C_i$ is convex.

“The intersection of (any number of) convex sets is convex.”

Example. *We consider the set*

$$S = \{x \in \mathbb{R}^m \mid |p(t)| \leq 1 \text{ for } |t| \leq \pi/3\},$$

where $p(t) = x_1 \cos(t) + \dots + x_m \cos(mt)$.

The set S can be expressed as the intersection of (convex) slabs.

$$S = \bigcap_{|t| \leq \pi/3} \{x \mid |p(t)| \leq 1\}.$$

Example. The convex cone $S_+^n \subseteq S^n$ can be expressed as

$$S_+^n = \bigcap_{z \neq 0} \{X \in S^n \mid z^T X z \geq 0\}.$$

For each $z \neq 0$, $z^T X z$ is a (not identically zero) linear function of X , so the sets

$$\{X \in S^n \mid z^T X z \geq 0\}$$

are, in fact, halfspaces in S^n . Thus the positive semidefinite cone is the intersection of an infinite number of halfspaces, and so is convex.

2.1.2 Affine mapping.

Suppose $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is affine, i.e., $f(x) = Ax + b$ with $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$.

Observation. $S \subseteq \mathbb{R}^n$ convex $\implies f(S) = \{f(x) \mid x \in S\}$ convex.

“The image of a convex set under f is convex.”

Observation. $C \subseteq \mathbb{R}^m$ convex $\implies f^{-1}(C) = \{x \in \mathbb{R}^n \mid f(x) \in C\}$ convex.

“The inverse image $f^{-1}(C)$ of a convex set under f is convex.”

Examples. If C is convex, then

- $\alpha \cdot C = \{\alpha \cdot x \mid x \in C\}$ is convex $\forall \alpha \in \mathbb{R}$.
- $a + C = \{a + x \mid x \in C\}$ is convex $\forall a \in \mathbb{R}^n$.
- C_1, C_2 convex in $\mathbb{R}^n \implies C_1 \times C_2$ is convex in $\mathbb{R}^{2n}$.
- C_1, C_2 convex in $\mathbb{R}^n \implies C_1 + C_2 = \{x + y \mid x \in C_1, y \in C_2\}$ is convex.

2.1.3 Perspective function.

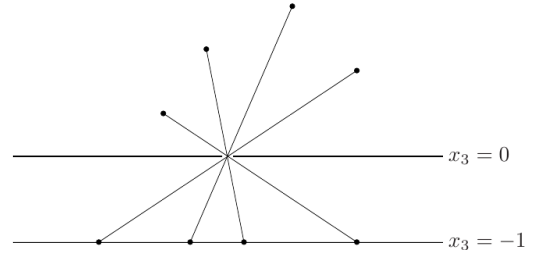
Perspective function $P : \mathbb{R}^{n+1} \rightarrow \mathbb{R}^n$ is defined as:

$$P(x, t) = x/t, \quad \text{dom } P = \{(x, t) \mid t > 0\}.$$

Images and inverse images of convex sets under perspective are convex:

$C \subseteq \mathbb{R}^n \times \mathbb{R}_+$ is convex $\implies P(C)$ is convex.

$C \subseteq \mathbb{R}^n \implies P^{-1}(C)$ is convex.



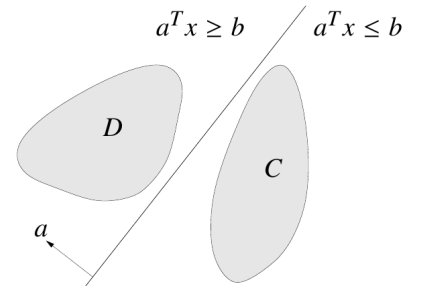
2.2 Separating hyperplane theorem.

Let $C, D \subseteq \mathbb{R}^n$ convex sets s.t. $C \cap D = \emptyset \implies \exists a \neq 0, b$ s.t.

$$a^T x \leq b \text{ for } x \in C, \quad a^T x \geq b \text{ for } x \in D$$

The hyperplane $\{x \mid a^T x = b\}$ separates C and D .

By additional additional assumptions we can define *strict separation*.



Definition (Distance). For $c \in C, d \in D$: $\text{dist}(C, D) = \inf \|c - d\|_2$.

Theorem (Strict separation (version 1)). Let $C, D \subseteq \mathbb{R}^n$ convex sets $C \cap D = \emptyset$, $\exists c \in C, d \in D$: $\text{dist}(C, D) = \|c - d\|_2 \implies \exists f$ affine such that

$$f(x) < 0, \forall x \in C, \quad f(y) > 0, \forall x \in D.$$

Proof. Let $f(x) = a^T x + b$ and $a = (d - c)$.

$$0 = f\left(\frac{c+d}{2}\right) \implies b = \frac{\|d\|_2^2 - \|c\|_2^2}{2} \rightsquigarrow \frac{1}{2}(d-c)^T(c+d) + b = 0$$

$$\begin{cases} f(c) = -\frac{1}{2}\|d-c\|_2^2 < 0 \\ f(d) = \frac{1}{2}\|d-c\|_2^2 > 0 \end{cases}$$

Hence for contradiction assume that $\exists u \in D : f(u) < 0$. Because $f(u) = (d-c)^T(u-d) + \frac{\|d-c\|_2^2}{2}$ and because $\frac{\|d-c\|_2^2}{2} > 0$ we can write just $(d-c)^T(u-d) < 0 \rightsquigarrow \|u + t(u-d) - c\|_2^2$

$$f(t) = \|d-c + t(u-d)\|_2^2 = \|d-c\|_2^2 + \underbrace{2t(d-c)^T(u-d)}_{<0} + t^2\|u-d\|_2^2$$

$$< \|d-c\|_2^2 \quad \text{for small enough } t > 0.$$

Hence $2(d-c)^T(u-d) + t\|u-d\|_2^2 < 0$ and therefore:

$$t < \frac{-2(d-c)^T(u-d)}{\|u-d\|_2^2} \text{ which is the contradiction.}$$

□

Definition (Open set). A set $U \subseteq \mathbb{R}^n$ is open if $\forall x \in U, \exists \varepsilon > 0$ such that

$$B(x, \varepsilon) = \{y \in \mathbb{R}^n \mid \|y - x\| < \varepsilon\} \subseteq U.$$

Definition (Closed set). A set $U \subseteq \mathbb{R}^n$ is closed $\iff \mathbb{R}^n \setminus U$ is open.

Definition (Compact set). A set is compact when it is closed and bounded.

Definition (Interior). Interior of $U = \text{int}(U) = \{x \in U \mid \exists \varepsilon > 0 : B(x, \varepsilon) \subseteq U\}$

Definition (Closure). Closure of U , $\text{clo}(U)$ is the smallest closed set containing U .

Definition (Boundary). Boundary of $U = \text{clo}(U) \setminus \text{int}(U)$.

Theorem (Strict separation (version 2)). Let $C, D \subseteq \mathbb{R}^n$ convex sets $C \cap D = \emptyset$, C compact and D closed $\implies \exists f$ affine such that

$$f(x) < 0, \forall x \in C, \quad f(y) > 0, \forall x \in D.$$

Lemma. Let $A \subseteq \mathbb{R}^n$ be compact and $B \subseteq \mathbb{R}^n$ be closed such that $A \cap B = \emptyset$. Then there exist $a \in A$ and $b \in B$ such that

$$\|a - b\|_2 = \text{dist}(A, B).$$

Proof. Let $a_0 \in A$, $b_0 \in B$, and define $r_1 = \|a_0 - b_0\| > 0$. Since A is compact, there exists $r_2 > 0$ such that $B_{r_2}(a_0) \supseteq A$. Define

$$S = B_{r_1+r_2}(a_0) \cap B \neq \emptyset.$$

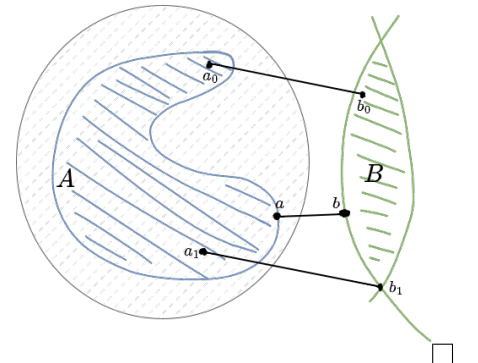
Then S is compact. Consider the continuous function now

$$A \times S \rightarrow \mathbb{R}, \quad (a, s) \mapsto \|a - s\|_2.$$

There exist $a' \in A$ and $b' \in S$ such that

$$\|a' - b'\| = \text{dist}(A, S) = \text{dist}(A, B).$$

Hence, there exist $a' \in A$ and $b' \in B$ with $\|a' - b'\|_2 = \text{dist}(A, B)$.

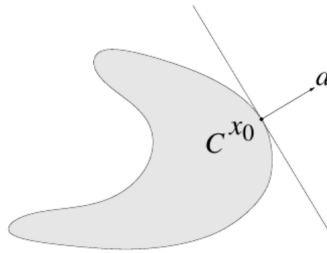


□

2.3 Supporting hyperplane theorem.

Suppose x_0 is a boundary point of set $C \subset \mathbb{R}^n$.

Definition (Supporting hyperplane). Supporting hyperplane to C at x_0 has form $\{x \mid a^T x = a^T x_0\}$, where $a \neq 0$ and $a^T x \leq a^T x_0$ for all $x \in C$.

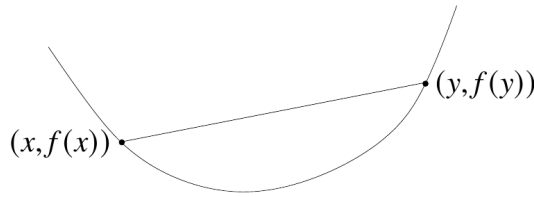


Theorem (Supporting hyperplane theorem). If C is convex, then there exists a supporting hyperplane at every boundary point of C .

3 Convex functions

Definition (Convex function). *Function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex if $\text{dom} f$ is a convex set and $\forall x, y \in \text{dom} f$, $0 \leq \theta \leq 1$:*

$$f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y)$$



Definition (Convex function). *Function f is concave if $-f$ is convex.*

Definition (Strictly convex function). *Function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is strictly convex if $\text{dom} f$ is convex and for $x, y \in \text{dom} f$, $x \neq y$, $0 \leq \theta \leq 1$:*

$$f(\theta x + (1 - \theta)y) < \theta f(x) + (1 - \theta)f(y)$$

Examples of convex/concave functions.

- Affine (convex + concave): $ax + b$ on $\mathbb{R}$, for any $a, b \in \mathbb{R}$.
- Exponential (convex): e^{ax} , for any $a \in \mathbb{R}$.
- Logarithm (concave): $\log x$ on $\mathbb{R}_{++}$.
- Powers (convex): x^α on $\mathbb{R}_{++}$, for $\alpha \geq 1$ or $\alpha \leq 0$, and concave for $0 \leq \alpha \leq 1$.
- Norms for $f : \mathbb{R}^n \rightarrow \mathbb{R}$ (convex):
 - $\|x\|_p = (|x_1|^p + \dots + |x_n|^p)^{1/p}$ for $p \geq 1$,
 - $\|x\|_\infty = \max\{|x_1|, \dots, |x_n|\}$.
- Max function for $f : \mathbb{R}^n \rightarrow \mathbb{R}$ (convex) length 31.80 km, difficulty 2.67/5 Runner 4: segments [6, 22, 33], $\max(x) = \max\{x_1, x_2, \dots, x_n\}$.
- Log-determinant: for $X \in S_{++}^n$, $f(X) = \log \det X$ is concave.

Example. *Show that $f(x_1, x_2) = x_1 \cdot x_2$ is convex, concave or neither.*

Proof. We first compute the gradient: $\nabla f = \begin{pmatrix} x_2 \\ x_1 \end{pmatrix}$. Then we compute the Hessian matrix and its determinant:

$$\nabla^2 f = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \rightsquigarrow \det \begin{pmatrix} -\lambda & 1 \\ 1 & -\lambda \end{pmatrix} = \lambda^2 - 1 = 0.$$

So $\lambda^2 = 1$ and therefore $\lambda = \pm 1$. So $\nabla^2 f$ is *indefinite* and is neither convex nor concave. □

3.1 First-order and Second-order conditions.

Definition. f is differentiable if $\text{dom } f$ is open and the gradient

$$\nabla f(x) = \begin{pmatrix} \frac{\partial f(x)}{\partial x_1} \\ \vdots \\ \frac{\partial f(x)}{\partial x_n} \end{pmatrix} \in \mathbb{R}^n \quad \text{is defined } \forall x \in \text{dom}.$$

Theorem (1st-order condition). Differentiable f with convex domain is convex $\iff$

$$f(y) \geq f(x) + \nabla f(x)^T(y - x), \quad \forall x, y \in \text{dom } f.$$

Proof. Consider $n = 1$ first. We show that a differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$ is convex $\iff$

$$f(y) \geq f(x) + f'(x)(y - x), \quad \forall x, y \in \text{dom } f$$

Assume first that f is convex and $x, y \in \text{dom } f$. Since $\text{dom } f$ is convex, we conclude that $\forall 0 < t \leq 1$, $x + t(y - x) \in \text{dom } f$, and by convexity of f ,

$$\begin{aligned} f(x + t(y - x)) &\leq (1 - t)f(x) + tf(y) \quad // \text{ divide both sides by } t \\ f(y) &\geq f(x) + \frac{f(x + t(y - x)) - f(x)}{t} \end{aligned}$$

and taking the limit as $t \rightarrow 0$ yields the formula.

To show sufficiency, assume the function satisfies the formula $\forall x, y \in \text{dom } f$. Choose any $x \neq y$, and $0 \leq \theta \leq 1$, and let $z = \theta x + (1 - \theta)y$. Applying (3.4) twice yields

$$f(x) \geq f(z) + f'(z)(x - z), \quad f(y) \geq f(z) + f'(z)(y - z).$$

Multiplying the first inequality by θ , the second by $1 - \theta$, and adding them yields

$$\theta f(x) + (1 - \theta)f(y) \geq f(z),$$

which proves that f is convex.

Now we can prove the general case, with $f : \mathbb{R}^n \rightarrow \mathbb{R}$. Let $x, y \in \mathbb{R}^n$ and consider f restricted to the line passing through them, i.e., the function defined by $g(t) = f(ty + (1 - t)x)$, so $g'(t) = \nabla f(ty + (1 - t)x)^T(y - x)$.

First assume f is convex, which implies g is convex, so by the argument above we have $g(1) \geq g(0) + g'(0)$, which means

$$f(y) \geq f(x) + \nabla f(x)^T(y - x).$$

Now assume that this inequality holds $\forall x, y$, so if $ty + (1 - t)x \in \text{dom } f$ and $\tilde{t}y + (1 - \tilde{t})x \in \text{dom } f$, we have

$$f(ty + (1 - t)x) \geq f(\tilde{t}y + (1 - \tilde{t})x) + \nabla f(\tilde{t}y + (1 - \tilde{t})x)^T(y - x)(t - \tilde{t}),$$

i.e., $g(t) \geq g(\tilde{t}) + g'(\tilde{t})(t - \tilde{t})$. We have seen that this implies that g is convex. $\square$

Definition. The function f is twice differentiable if $\text{dom } f$ is open and the Hessian $\nabla^2 f(x) \in S^n$,

$$\nabla^2 f(x)_{ij} = \frac{\partial^2 f(x)}{\partial x_i \partial x_j}, \quad i, j = 1, \dots, n,$$

exists at each $x \in \text{dom } f$.

Theorem (2nd-order conditions). *For twice differentiable f with convex domain*

- f is convex $\iff \nabla^2 f(x) \succeq 0, \quad \forall x \in \text{dom} f,$
- if $\nabla^2 f(x) \succ 0, \quad \forall x \in \text{dom} f,$ then f is strictly convex.

Proof. Let $\xi \in [x, y]$. For all x, y :

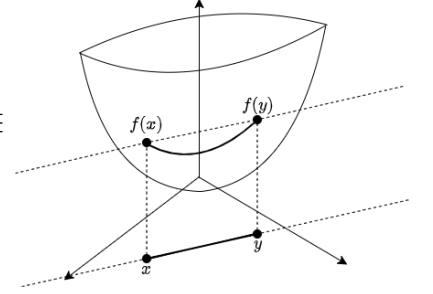
$$\begin{aligned} f(y) &= f(x) + \nabla f(x)^T(y - x) + \frac{1}{2}(y - x)^T \underbrace{\nabla^2 f(\xi)}_{\succeq 0}(y - x) \\ &\geq f(x) + \nabla f(x)^T(y - x), \end{aligned}$$

□

Example. *Function $f(x) = -x \log x$ is strictly convex:*

$$f'(x) = -\log x - 1, \quad f''(x) = -\frac{1}{x} < 0$$

Observation. *The function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex if $\forall x \in \mathbb{R}^n, u \in \mathbb{R}^n \setminus \{0\} : g(t) = f(x + tu)$ is convex.*



Example (Log-determinant). *The function $f(X) = \log \det X$ is concave on $\text{dom } f = S_{++}^n$.*

Proof. We can verify concavity by considering an arbitrary line, given by $X = Z + tV$, where $Z, V \in S^n$. We define $g(t) = f(Z + tV)$, and restrict g to the interval of values of t for which $Z + tV \succ 0$. Without loss of generality, we can assume that $t = 0$ is inside this interval, i.e., $Z \succ 0$. We have

$$\begin{aligned} g(t) &= \log \det(Z + tV) \\ &= \log \det(Z^{1/2}(I + tZ^{-1/2}VZ^{-1/2})Z^{1/2}) \\ &= \sum_{i=1}^n \log(1 + t\lambda_i) + \log \det Z \end{aligned}$$

where $\lambda_1, \dots, \lambda_n$ are the eigenvalues of $Z^{-1/2}VZ^{-1/2}$. Therefore we have

$$g'(t) = \sum_{i=1}^n \frac{\lambda_i}{1 + t\lambda_i}, \quad g''(t) = -\sum_{i=1}^n \frac{\lambda_i^2}{(1 + t\lambda_i)^2}.$$

Since $g''(t) \leq 0$, we conclude that f is concave.

□

Example (Quadratic function).

$$f(x) = (1/2)x^T Px + q^T x + r \quad (\text{with } P \in S^n)$$

$$\nabla f(x) = Px + q, \quad \nabla^2 f(x) = P$$

convex if $P \succeq 0$ (concave if $P \succ 0$).

Example (Softmax or log-sum-exp function). Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ such that

$$f(x) = \log(e^{x_1} + e^{x_2} + \cdots + e^{x_n}) \quad \text{is convex.}$$

First we notice:

$$\max(x) \leq \log(e^{x_1} + e^{x_2} + \cdots + e^{x_n}) \leq \log(ne^{\max(x)}) = \log n + \max(x)$$

We compute the first derivative:

$$\frac{\partial}{\partial x_i} f(x) = \frac{e^{x_i}}{e^{x_1} + \cdots + e^{x_n}}, \quad \nabla f(x) = \frac{1}{\sum e^{x_j}} \begin{pmatrix} e^{x_1} \\ \vdots \\ e^{x_n} \end{pmatrix}, \quad \text{and let } z_i = e^{x_i}.$$

Now we compute the Hessian matrix

$$\nabla^2 f(x) = \frac{1}{\sum_{i=1}^n z_i} \begin{pmatrix} z_1 & z_2 & \cdots & z_n \\ z_2 & z_2^2 & \cdots & z_2 z_n \\ \vdots & \vdots & \ddots & \vdots \\ z_n & z_n z_2 & \cdots & z_n^2 \end{pmatrix} - \frac{1}{(\sum_{i=1}^n z_i)^2} \begin{pmatrix} z_1 z_1 & z_1 z_2 & \cdots & z_1 z_n \\ z_2 z_1 & z_2 z_2 & \cdots & z_2 z_n \\ \vdots & \vdots & \ddots & \vdots \\ z_n z_1 & z_n z_2 & \cdots & z_n z_n \end{pmatrix}$$

Which we can rewrite as follows:

$$\nabla^2 f(z) = \frac{1}{(\mathbf{1}^T z)} \text{diag}(z) - \frac{1}{(\mathbf{1}^T z)^2} z z^T, \quad \text{where; } \mathbf{1} = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix}, \quad z = \begin{pmatrix} z_1 \\ \vdots \\ z_n \end{pmatrix} \in \mathbb{R}_+^n$$

Check for all $v \in \mathbb{R}^n$:

$$\begin{aligned} 0 \leq v^T \nabla^2 f(z) v &= \frac{1}{(\mathbf{1}^T z)^2} \left[\left(\sum_{k=1}^n z_k v_k^2 \right) \left(\sum_{k=1}^n z_k \right) - \left(\sum_{k=1}^n v_k z_k \right)^2 \right] \\ &= \frac{1}{(\mathbf{1}^T z)^2} \left[\left\| \begin{pmatrix} v_1 \sqrt{z_1} \\ \vdots \\ v_n \sqrt{z_n} \end{pmatrix} \right\|^2 \left\| \begin{pmatrix} \sqrt{z_1} \\ \vdots \\ \sqrt{z_n} \end{pmatrix} \right\|^2 - \left(\begin{pmatrix} v_1 \sqrt{z_1} \\ \vdots \\ v_n \sqrt{z_n} \end{pmatrix}^T \begin{pmatrix} \sqrt{z_1} \\ \vdots \\ \sqrt{z_n} \end{pmatrix} \right)^2 \right] \end{aligned}$$

And by the Cauchy-Schwarz inequality, $(a, b)^2 \leq \|a\|^2 \|b\|^2$, so $v^T \nabla^2 f(z) v \geq 0$.

3.2 Epigraph and sublevel set.

Definition (α -sublevel). α -sublevel set of $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is

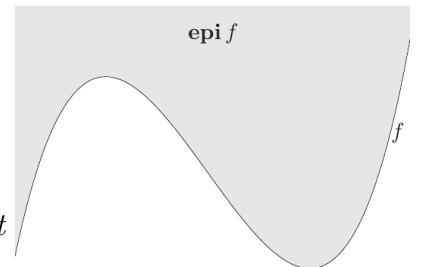
$$C_\alpha = \{x \in \text{dom } f \mid f(x) \leq \alpha\}.$$

Definition (Epigraph). Epigraph of $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is

$$\text{epi } f = \{(x, t) \in \mathbb{R}^{n+1} \mid x \in \text{dom } f, f(x) \leq t\}.$$

Observation. Sublevel sets of convex functions are convex sets (but converse is false).

Observation. The function f is convex $\iff$ $\text{epi } f$ is a convex set.



3.3 Showing a function is convex

Methods for establishing convexity of a function f :

1. Verify definition (often simplified by restricting to a line).
2. For twice differentiable functions, show $\nabla^2 f(x) \geq 0$.
 - recommended only for very simple functions.
3. Show that f is obtained from simple convex functions by operations that preserve convexity
 - nonnegative weighted sum
 - composition with affine function
 - pointwise maximum and supremum
 - composition
 - minimization
 - perspective

Theorem (Jensen's inequality). *If f is convex, then for $x, y \in \text{dom } f$, $0 \leq \theta \leq 1$,*

$$f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y)$$

Nonnegative scaling, sum, and integral.

- **Nonnegative multiple:** αf is convex if f is convex and $\alpha \geq 0$.
- **Sum:** $f_1 + f_2$ convex if f_1, f_2 convex.
- **Infinite sum:** If $f_1, f_2, \dots$ are convex functions, infinite sum $\sum_{i=1}^{\infty} f_i$ is convex
- **Integral:** If $f(x, \alpha)$ is convex in x for each $\alpha \in \mathcal{A}$, then

$$\int_{\alpha \in \mathcal{A}} f(x, \alpha) d\alpha \quad \text{is convex.}$$

Composition with affine function.

Observation. *Function $f(Ax + b)$ is convex if f is convex.*

Example. *Log barrier for linear inequalities:*

$$f(x) = - \sum_{i=1}^m \log(b_i - a_i^T x), \quad \text{dom } f = \{x \mid a_i^T x < b_i, i = 1, \dots, m\}.$$

Pointwise maximum.

Observation. *If $f_1, \dots, f_m$ are convex, then $f(x) = \max\{f_1(x), \dots, f_m(x)\}$ is convex.*

Proof. We know that f is convex $\iff \text{epi}(f)$ is convex. By that:

$$\begin{aligned} (x, t) \in \text{epi}(f) &\iff f(x) \leq t \iff \forall i : f_i(x) \leq t \\ &\iff \forall i : (x, t) \in \text{epi}(f_i) \iff \text{epi}(f) = \bigcap_{i=1}^m \text{epi}(f_i). \end{aligned}$$

□

Example. *Piecewise-linear function $f(x) = \max_{i=1, \dots, m} (a_i^T x + b_i)$ is convex.*

Example. *Sum of r largest components of $x \in \mathbb{R}^n$:*

$$f(x) = x_{[1]} + \dots + x_{[r]}, \quad \text{where } x_{[i]} \text{ is } i\text{-th largest component of } x, \text{ is convex.}$$

Proof: $f(x) = \max\{x_{i_1} + x_{i_2} + \dots + x_{i_r} \mid 1 \leq i_1 < i_2 < \dots < i_r \leq n\}$

Pointwise supremum.

Observation. If $f(x, y)$ is convex in x for each $y \in \mathcal{A}$, then $g(x) = \sup_{y \in \mathcal{A}} f(x, y)$ is convex.

Example. Distance to farthest point in a set C : $f(x) = \sup_{y \in C} \|x - y\|_2$ is convex.

Partial minimization.

Definition (Partial minimization). The function $g(x) = \inf_{y \in C} f(x, y)$ is called the partial minimization of f (w.r.t. y).

Observation. If $f(x, y)$ is convex in (x, y) and C is a convex set, then partial minimization g is convex.

Example. Distance to a set: $\text{dist}(x, S) = \inf_{y \in S} \|x - y\|$ is convex if S is convex.

Composition with scalar functions.

Definition (Composition). Composition of $g : \mathbb{R}^n \rightarrow \mathbb{R}$ and $h : \mathbb{R} \rightarrow \mathbb{R}$ is $f(x) = h(g(x))$ (w.a. $f = h \circ g$).

Observation. Composition f is convex if

- g convex, h convex, $\tilde{h}$ nondecreasing
- or g concave, h convex, $\tilde{h}$ nonincreasing

Example. For $n = 1$, differentiable g, h :

$$\begin{aligned} f'(x) &= h'(g(x))g'(x) \\ f''(x) &= \underbrace{h''(g(x))}_{\geq 0} \underbrace{g'(x)^2}_{\geq 0} + \underbrace{h'(g(x))}_{\geq 0} \underbrace{g''(x)}_{\geq 0}. \end{aligned}$$

Example. If f is convex then $e^{g(x)}$ is convex.

Example. If f is convex then $\frac{1}{g(x)}$ is convex.

General composition rule.

Definition (General composition). Composition of $g : \mathbb{R}^n \rightarrow \mathbb{R}^k$ and $h : \mathbb{R}^k \rightarrow \mathbb{R}$ is $f(x) = h(g(x)) = h(g_1(x), g_2(x), \dots, g_k(x))$

Observation. f is convex if h is convex and for each i one of the following holds

- g_i convex, $\tilde{h}$ nondecreasing in its i th argument
- g_i concave, $\tilde{h}$ nonincreasing in its i th argument
- g_i affine

Example. If $g_1, \dots, g_n$ are convex then $f(x) = \log \left(\sum_i e^{g_i(x)} \right)$ is convex.

Perspective.

Definition (Perspective). The perspective of a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is the function $g : \mathbb{R}^n \times \mathbb{R} \rightarrow \mathbb{R}$,

$$g(x, t) = tf\left(\frac{x}{t}\right), \quad \text{dom } g = \left\{(x, t) \mid \frac{x}{t} \in \text{dom } f, t > 0\right\}$$

Observation. Function f is convex $\implies g$ is convex.

Example. $f(x) = x^T x$ is convex; so $g(x, t) = t \frac{x^T x}{t} = \frac{x^T x}{t}$ is convex for $t > 0$.

Proof.

$$\begin{aligned} (x, t, s) \in \text{epi}(g) &\iff t \cdot f\left(\frac{x}{t}\right) \leq s \iff \\ &\iff f\left(\frac{x}{t}\right) \leq \frac{s}{t} \iff \left(\frac{x}{t}, \frac{s}{t}\right) \in \text{epi}(f), \quad \text{is convex.} \end{aligned}$$

□

3.4 Quasiconvex functions

Definition (Quasiconvex function). The function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is quasiconvex if $\text{dom } f$ is convex and the sublevel sets are convex for all α :

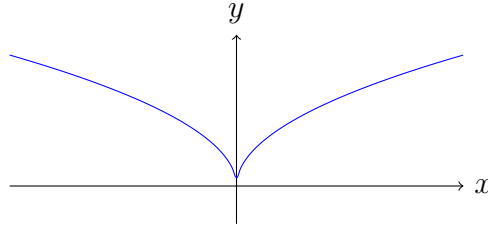
$$S_\alpha = \{x \in \text{dom } f \mid f(x) \leq \alpha\}$$

Definition (Quasiconcave function). The function f is quasiconcave if $-f$ is quasiconvex.

Definition (Quasilinear function). The function f is quasilinear if it is quasiconvex and quasiconcave.

Example. $\text{ceil}(x) = \inf\{z \in \mathbb{Z} \mid z \geq x\}$ is quasilinear.

Example. $\sqrt{|x|}$ is quasiconvex on $\mathbb{R}$.



Example. Linear-fractional function as below is quasilinear.

$$f(x) = \frac{a^T x + b}{c^T x + d}, \quad \text{dom } f = \{x \mid c^T x + d > 0\}.$$

Example (Internal rate of return (IRR)). We have cash flow $x = (x_0, \dots, x_n)$; x_i is payment in period i (to us if $x_i > 0$). We assume an initial investment $x_0 < 0$ and $x_0 + x_1 + \dots + x_n > 0$.

- **Net present value** (NPV) of cash flow x , for interest rate r , is

$$PV(x, r) = \sum_{i=0}^n (1+r)^{-i} x_i$$

- **Internal rate of return** (IRR) is smallest interest rate for which $PV(x, r) = 0$:

$$IRR(x) = \inf\{r \geq 0 \mid PV(x, r) = 0\}$$

- IRR is quasiconcave: superlevel set is intersection of open halfspaces

$$IRR(x) \geq R \iff \sum_{i=0}^n (1+r)^{-i} x_i > 0 \text{ for } 0 \leq r < R$$

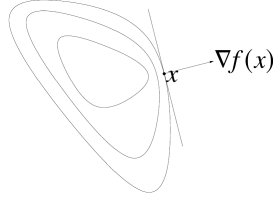
Properties of quasiconvex functions.

Theorem (Modified Jensen inequality). *For quasiconvex f*

$$0 \leq \theta \leq 1 \implies f(\theta x + (1 - \theta)y) \leq \max\{f(x), f(y)\}.$$

Theorem (First-order condition). *Differentiable f with convex domain is quasiconvex iff*

$$f(y) \leq f(x) \implies \nabla f(x)^T(y - x) \leq 0.$$



Observation. *The sum of quasiconvex functions is not necessarily quasiconvex.*

4 Optimization problem in standard form.

$$\begin{aligned} & \text{minimize} && f_0(x) \\ & \text{subject to} && f_i(x) \leq 0, \quad i = 1, \dots, m \\ & && h_i(x) = 0, \quad i = 1, \dots, p \end{aligned}$$

- $x \in \mathbb{R}^n$ is the optimization variable
- $f_0 : \mathbb{R}^n \rightarrow \mathbb{R}$ is the objective or cost function
- $f_i : \mathbb{R}^n \rightarrow \mathbb{R}, i = 1, \dots, m$, are the inequality constraint functions
- $h_i : \mathbb{R}^n \rightarrow \mathbb{R}$ are the equality constraint functions

Definition (Feasible point). *Point $x \in \mathbb{R}^n$ is feasible if $x \in \text{dom } f_0$ and it satisfies the constraints. The problem is infeasible if $p^* = \infty$ and unbounded if $p^* = -\infty$.*

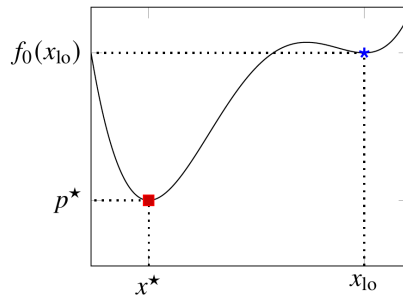
Definition (Optimal value). *Optimal value is*

$$p^* = \inf\{f_0(x) \mid f_i(x) \leq 0, i = 1, \dots, m, h_i(x) = 0, i = 1, \dots, p\}.$$

A feasible x is optimal if $f_0(x) = p^$.*

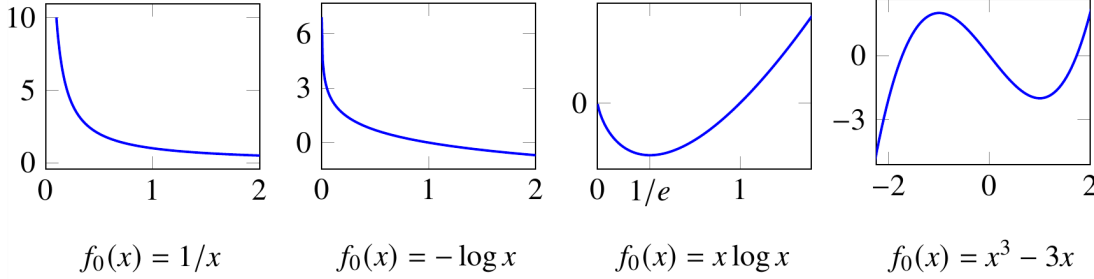
X_{opt} is the set of optimal points.

Definition (Locally optimal point). *The point x is locally optimal if there is an $R > 0$ such that x is optimal for a standard optimization problem $+ \|z - x\|_2 \leq R$.*



Example. Examples with $n = 1$, $m = p = 0$:

- $f_0(x) = 1/x$, $\text{dom}f_0 = \mathbb{R}_{++}$: $p^* = 0$, no optimal point
- $f_0(x) = -\log x$, $\text{dom}f_0 = \mathbb{R}_{++}$: $p^* = -\infty$
- $f_0(x) = x \log x$, $\text{dom}f_0 = \mathbb{R}_{++}$: $p^* = -1/e$, $x = 1/e$ is optimal
- $f_0(x) = x^3 - 3x$: $p^* = -\infty$, $x = 1$ is locally optimal



Observation. Feasible and optimal sets of a convex optimization problem are convex.

Proof. Let $X := \{\text{feasible points}\}$.

$$X = \underbrace{\text{dom}(f_0)}_{\text{convex}} \cap \bigcap_{i=1}^m \underbrace{\{x \mid f_i(x) \leq 0\}}_{\text{convex}} \cap \bigcap_{i=1}^p \underbrace{\{x \mid h_i(x) = 0\}}_{\text{convex}}.$$

□

Theorem. Suppose that the objective f_0 in a cvxopt is differentiable, so that $\forall x, y \in \text{dom } f_0$,

$$f_0(y) \geq f_0(x) + \nabla f_0(x)^T(y - x)$$

Let X denote the feasible set. Then x is optimal $\iff x \in X$ and

$$\nabla f_0(x)^T(y - x) \geq 0, \quad \forall y \in X.$$

Proof. First suppose $x \in X$ and satisfies $\nabla f_0(x)^T(y - x) \geq 0$. Then if $y \in X$ we have, by differentiability, $f_0(y) \geq f_0(x)$. This shows x is an optimal point.

Conversely, suppose x is optimal, but the condition does not hold, i.e., for some $y \in X$ we have

$$\nabla f_0(x)^T(y - x) < 0.$$

Consider the point $z(t) = ty + (1 - t)x$, where $t \in [0, 1]$ is a parameter. Since $z(t)$ is on the line segment between x and y , and the feasible set is convex, $z(t)$ is feasible. We claim that for small positive t we have $f_0(z(t)) < f_0(x)$, which will prove that x is not optimal. To show this, note that

$$\left. \frac{d}{dt} f_0(z(t)) \right|_{t=0} = \nabla f_0(x)^T(y - x) < 0,$$

so for small positive t , we have $f_0(z(t)) < f_0(x)$. □

Theorem. Any locally optimal point x_{loc} of a convex problem is (globally) optimal.

Proof. Suppose x_{loc} is locally optimal, but there exists a feasible y with $f_0(y) < f_0(x_{\text{loc}})$. Since x_{loc} is locally optimal, there $\exists R > 0$ s.t. for any feasible z with $\|z - x_{\text{loc}}\|_2 \leq R$, we have $f_0(z) \geq f_0(x_{\text{loc}})$.

Now consider $z = \theta y + (1 - \theta)x_{\text{loc}}$ with $\theta = R/(2\|y - x_{\text{loc}}\|_2)$. Since $\|y - x_{\text{loc}}\|_2 > R$, we have $0 < \theta < 1/2$. The point z is a convex combination of two feasible points and is therefore also feasible. We have $\|z - x_{\text{loc}}\|_2 = R/2$ and

$$f_0(z) \leq \theta f_0(y) + (1 - \theta)f_0(x_{\text{loc}}) < f_0(x_{\text{loc}}),$$

which contradicts our assumption that x_{loc} is locally optimal. □

Observation (Unconstrained problem). x minimizes $f_0(x) \iff \nabla f_0(x) = 0$.

Programs.

Definition (Linear Program (LP)).

$$\begin{array}{ll} \text{minimize} & c^T x + d \\ \text{subject to} & Gx \leq h \\ & Ax = b \end{array}$$

The objective and constraint functions are all affine. The feasible set is a polyhedron. Example: Diet problem (minimize cost subject to nutrient constraints).

Definition (Quadratic Program (QP)).

$$\begin{array}{ll} \text{minimize} & (1/2)x^T P x + q^T x + r \\ \text{subject to} & Gx \leq h \\ & Ax = b \end{array}$$

where $P \in \mathbf{S}_+^n$ (positive semidefinite). The objective is convex quadratic, constraints are affine. The feasible set is a polyhedron. Example: Least-squares problem with linear constraints.

Definition (Quadratically Constrained Quadratic Program (QCQP)).

$$\begin{array}{ll} \text{minimize} & (1/2)x^T P_0 x + q_0^T x + r_0 \\ \text{subject to} & (1/2)x^T P_i x + q_i^T x + r_i \leq 0, \quad i = 1, \dots, m \\ & Ax = b \end{array}$$

where $P_i \in \mathbf{S}_+^n$. The objective and inequality constraints are convex quadratic. The feasible set is the intersection of ellipsoids (if $P_i \succ 0$).

Dimension and Geometry of Polyhedra.

Definition (Affine Dimension). *The dimension of a set $C \subseteq \mathbb{R}^n$ is defined as the dimension of its affine hull:*

$$\dim(C) = \dim(\text{aff } C)$$

Example: The affine dimension of the unit circle in $\mathbb{R}^2$ is 2, but intrinsic dimension is 1.

Definition (Vertex – Unique Maximum). *Mathematically, a vertex v of a convex polyhedron P is defined as a point where some linear function $c^T x$ attains a unique maximum over P .*

Definition (Face). *A subset F of a convex set C is a face of C if it is the intersection of C with a supporting hyperplane. A vertex is a 0-dimensional face. An edge is a 1-dimensional face.*

Observation. *If a polyhedron $P = \{x | Ax \leq b\}$ is bounded (a polytope), then it is equal to the convex hull of its vertices:*

$$P = \text{conv}\{v_1, \dots, v_k\}$$

where $v_1, \dots, v_k$ are the vertices of P .

Proof idea. By induction on the dimension. If $x \in P$ is not a vertex, it lies on a line segment contained in P . We can extend this line segment in both directions until it hits the boundary (faces of lower dimension). By induction, the boundary points can be written as convex combinations of vertices. Therefore, x (a convex combination of boundary points) is a convex combination of vertices. Alternatively, related to the fact that a linear function on a bounded polyhedron attains its maximum at a vertex. $\square$

Standard Normal Form and The Simplex Method

Definition (Standard Normal Form (SNF)). *A linear program is in Standard Normal Form if it is a minimization problem with equality constraints and non-negative variables:*

$$\begin{aligned} & \text{maximize} && c^T x \\ & \text{subject to} && Ax = b \\ & && x \geq 0 \end{aligned}$$

where $A \in \mathbb{R}^{m \times n}$ has rank m (typically $m < n$), $b \in \mathbb{R}^m$, and $c \in \mathbb{R}^n$.

Observation. Every LP is equivalent to an LP in Standard Normal Form.

Transformations to SNF: In case LP is not in SNF, we follow these transformation rules :

- *Inequality to Equality:* For a constraint $a^T x \leq \beta$, introduce a *slack variable* $s \geq 0$:

$$a^T x + s = \beta, \quad s \geq 0$$

- *Unrestricted Variable:* If x is free (unbounded), substitute

$$x = x^+ - x^-, \quad \text{where } x^+ \geq 0, x^- \geq 0.$$

- *Minimization:* $\min c^T x \iff \max(-c)^T x$.

Definition (Basis matrix). Let $B \subseteq \{1, \dots, n\}$ be a set of m indices. We define the basis matrix A_B consisting of columns of A indexed by B :

$$A_B = [A_{\cdot j} \mid j \in B]$$

Definition (Basis). A set B is a basis if A_B is nonsingular (invertible).

Definition (Basic solution). A vector $x \in \mathbb{R}^n$ is a basic solution for basis B if $x_N = 0$ (for $N = \{1, \dots, n\} \setminus B$) and $x_B = A_B^{-1}b$.

Definition (Basic Feasible Solution (BFS)). A vector x is a Basic Feasible Solution (BFS) if it is a basic solution and $x \geq 0$ (i.e., $A_B^{-1}b \geq 0$).

Observation (Uniqueness). If a basic feasible solution exists for a fixed basis B , it is unique (determined by $x_B = A_B^{-1}b$).

Theorem. Let $P = \{x \mid Ax = b, x \geq 0\}$. A point $x \in P$ is a vertex of P if and only if x is a basic feasible solution of P .

Proof. Let $v \in P$ and $K = \{i \in \{1, \dots, n\} \mid v_i > 0\}$.

- *Case 1:* Columns A_K are linearly independent. Then we can extend indices K to a basis B (size m). Since v is uniquely determined by constraints on K and zeros elsewhere, it corresponds to a BFS.
- *Case 2:* Columns A_K are **not** linearly independent. Then there exists a vector $w_K \neq 0$ such that $A_K w_K = 0$. Define $w \in \mathbb{R}^n$ extending w_K with zeros. Then $Aw = 0$. We can form $v' = v + \epsilon w$ and $v'' = v - \epsilon w$. For small ϵ , both are feasible (non-negative). Since $v = \frac{1}{2}(v' + v'')$, v is not a vertex.

□

Observation. If v is a BFS with respect to basis B , it can be viewed as the optimum of:

$$\text{minimize } \sum_i x_i \quad \text{subject to original constraints (context dependent).}$$

The Simplex Method.

The Simplex method moves from one BFS to an adjacent BFS with a better objective value.

Definition (Simplex Tableau). *Given a basis B , we express basic variables x_B and the objective z in terms of non-basic variables x_N :*

$$\begin{aligned} x_B &= A_B^{-1}b - A_B^{-1}A_Nx_N \\ z &= c_B^T A_B^{-1}b + (c_N^T - c_B^T A_B^{-1}A_N)x_N \end{aligned}$$

The vector $r = c_N^T - c_B^T A_B^{-1}A_N$ represents the **reduced costs**.

Algorithm – Simplex method. The simplex method follows these steps:

1. *Check optimality:* If all reduced costs ≤ 0 (for maximization), current solution is optimal.
2. *Pivot Column (Entering Variable):* Choose $j \in N$ with reduced cost > 0 .
3. *Pivot Row (Leaving Variable):* Perform the *ratio test* to ensure feasibility ($x \geq 0$). Select $i \in B$ that minimizes the ratio x_i/a_{ij} for $a_{ij} > 0$.
4. *Update:* Perform Gaussian elimination to update the tableau for the new basis.

Example. We have a problem:

$$\begin{aligned} \text{maximize} \quad & x_1 + x_2 \\ \text{subject to} \quad & 2x_1 + x_2 \leq 4 \\ & x_1 + 2x_2 \leq 3 \\ & x_1, x_2 \geq 0 \end{aligned}$$

- *Step 1: Initialization (Slack Variables).* We introduce slack variables $x_3, x_4 \geq 0$ to convert to equations. Initial Basis $B = \{3, 4\}$, Non-basic $N = \{1, 2\}$. We express basic variables and objective z using non-basic variables:

$$\begin{array}{rcl} x_3 & = & 4 - 2x_1 - x_2 \\ x_4 & = & 3 - x_1 - 2x_2 \\ \hline z & = & 0 + x_1 + x_2 \end{array}$$

Current BFS: $x_1 = 0, x_2 = 0 \implies x_3 = 4, x_4 = 3$. Objective $z = 0$.

- *Step 2: First Pivot.*
 - *Entering Variable:* x_1 (coefficient in z is $+1 > 0$).
 - *Leaving Variable (Ratio Test):* Constraint 1: $4/2 = 2$ (bound for x_1). Constraint 2: $3/1 = 3$. Minimum is 2, so x_3 leaves the basis.

Rearrange first row to express x_1 : $x_1 = 2 - 0.5x_2 - 0.5x_3$. Substitute into others.

- *Step 3: Second Dictionary.* New Basis $B = \{1, 4\}$.

$$\begin{array}{rcl} x_1 & = & 2 - \frac{1}{2}x_2 - \frac{1}{2}x_3 \\ x_4 & = & 3 - (2 - \frac{1}{2}x_2 - \frac{1}{2}x_3) - 2x_2 = 1 - \frac{3}{2}x_2 + \frac{1}{2}x_3 \\ \hline z & = & (2 - \frac{1}{2}x_2 - \frac{1}{2}x_3) + x_2 = 2 + \frac{1}{2}x_2 - \frac{1}{2}x_3 \end{array}$$

Current BFS: $x_2 = 0, x_3 = 0 \implies x_1 = 2, x_4 = 1$. Objective $z = 2$.

- *Step 4: Second Pivot.*
 - *Entering Variable:* x_2 (coefficient in z is $+\frac{1}{2} > 0$).

– *Leaving Variable:* Row x_1 : $2/\frac{1}{2} = 4$. Row x_4 : $1/\frac{3}{2} = \frac{2}{3}$. Minimum is $\frac{2}{3}$, so x_4 leaves.

- *Step 5: Final Dictionary* Rearrange row x_4 to express x_2 : $x_2 = \frac{2}{3} + \frac{1}{3}x_3 - \frac{2}{3}x_4$. Substitute into z (and x_1).

$$z = 2 + \frac{1}{2}\left(\frac{2}{3} + \dots\right) - \frac{1}{2}x_3 = \frac{7}{3} - \dots x_3 - \dots x_4$$

Exact optimal value: $z = 7/3$. Since all coefficients in z row for non-basic variables are negative, we terminate.

Possible Problems with Simplex method.

Unboundedness.

Occurs when the entering variable can be increased indefinitely because it has non-positive coefficients in all constraint rows of the dictionary. Such as:

$$\begin{array}{rcl} x_3 & = & 5 + x_1 - x_2 \\ x_4 & = & 2 + 2x_1 + x_2 \\ \hline z & = & 10 + 3x_2 \end{array}$$

Here, we want to increase x_2 (coeff +3). Looking at constraints:

- Row 1: coefficient of x_2 is -1 . As $x_2 \uparrow$, x_3 increases (no limit).

- Row 2: coefficient of x_2 is $+1$ (but sign in eq is usually minus, here $x_4 = 2 + x_2$

implies $x_4 - x_2 = 2$, this notation assumes $x_B = b - A_N x_N$. If strictly following standard notation: $x_4 = 2 + x_2$ means no constraint limits x_2).

Simpler Geometric Example: $\max x_1$ subject to $x_1 \geq 0, x_2 \geq 0, x_1 - x_2 \leq 1$. We can increase x_1 and x_2 simultaneously forever.

Degeneracy.

Occurs when a basic variable has a value of 0. This happens if there is a "tie" in the ratio test during a previous step. Geometrically, the vertex is over-determined (more than n hyperplanes intersect). Such as

$$\begin{array}{rcl} x_3 & = & 0 - x_1 + x_2 \\ x_4 & = & 2 - x_1 - x_2 \\ \hline z & = & 5 + x_1 \end{array}$$

Here, basic variable $x_3 = 0$. If we pivot on x_1 , the ratio for x_3 is $0/1 = 0$. The step length is 0. The basis changes (x_1 enters, x_3 leaves), but the solution x and objective value z remain unchanged.

Cycling.

Occurs when the algorithm moves through a sequence of degenerate bases and returns to the starting basis. This leads to an infinite loop.

- *Condition:* Requires Degeneracy + Specific Pivot Rule (e.g., always choosing the first eligible column).
- *Counter-measure:* Bland's Rule (choose smallest index for entering/leaving variable) prevents cycling.
- *Famous Example:* Marshall and Kuhn (1951) constructed a specific dictionary with $n = 4, m = 3$ that cycles in 6 pivots if standard rules are used.

Finding a Basic Feasible Solution (Phase I).

The Simplex method requires an initial Basic Feasible Solution to start. If the origin $x = 0$ is not feasible (e.g., constraints $Ax = b$ or $Ax \leq b$ where $b \not\geq 0$), we use the *Two-Phase Method*.

Phase I formulation. Construct an auxiliary LP to find a feasible point for the original problem. Given constraints $Ax = b$ with $b \geq 0$:

$$\begin{array}{ll}\text{minimize} & \sum_{i=1}^m w_i \\ \text{subject to} & Ax + Iw = b \\ & x \geq 0, w \geq 0\end{array}$$

where $w = (w_1, \dots, w_m)$ are *artificial variables*.

- **Initialization:** This auxiliary problem has a trivial BFS: $x = 0, w = b$.
- **Outcome:** Solve Phase I. Let the optimal objective value be z^* .
 1. If $z^* > 0$: The original problem is *infeasible*.
 2. If $z^* = 0$: The current solution has $w = 0$ and provides a BFS for the original problem (Phase II starts here).

Pivot Rules.

The pivot rule determines which entering variable (column) to choose when multiple reduced costs are positive.

- **Dantzig's Rule (Largest Coefficient):** Choose the variable with the largest positive reduced cost. (Standard, but can be slow).
- **Bland's Rule (Smallest Index):** Choose the eligible variable with the smallest index. Guarantees termination and prevents cycling.
- **Largest Increase:** Choose the variable that yields the maximal increase in the objective function value in the current step (requires more computation per step).
- **Steepest Edge:** Choose the direction that is geometrically "steepest" in the space of variables. Very efficient in terms of iteration count, but computationally expensive update.
- **Random Edge:** Choose an entering variable uniformly at random among eligible ones. (Randomized pivot rules are important for theoretical bounds).

LP Relaxations and Integer Programming.

Many combinatorial problems can be formulated as Integer Linear Programs (ILP). An *LP relaxation* is obtained by removing the integrality constraint.

Example (Vertex cover). *Given a graph $G = (V, E)$, find the minimum number of vertices to cover all edges.*

ILP

$$\begin{array}{ll}\text{minimize} & \sum_{v \in V} x_v \\ \text{subject to} & x_u + x_v \geq 1 \quad \forall \{u, v\} \in E \\ & x_v \in \{0, 1\} \quad \forall v \in V\end{array}$$

LP Relaxation

Replace $x_v \in \{0, 1\}$ with linear constraints $0 \leq x_v \leq 1$.

$$\begin{array}{ll}\text{minimize} & \sum_{v \in V} x_v \\ \text{subject to} & x_u + x_v \geq 1 \quad \forall \{u, v\} \in E \\ & 0 \leq x_v \leq 1\end{array}$$

Solving the LP relaxation gives a lower bound on the optimal integer value. For Vertex Cover, the LP solution is "half-integral" ($x_i \in \{0, \frac{1}{2}, 1\}$).

Linear-Fractional Programming.

Definition (Linear-fractional program). A linear-fractional program is given by:

$$\begin{array}{ll}\text{minimize} & f_0(x) = \frac{c^T x + d}{e^T x + f} \\ \text{subject to} & Gx \leq h \\ & Ax = b\end{array}$$

where the domain is $\{x | e^T x + f > 0\}$. The objective function is quasiconvex (and quasiconcave).

We can transform this problem into an equivalent LP using the **Charnes-Cooper transformation**. Introduce variables $y \in \mathbb{R}^n$ and $t \in \mathbb{R}$:

$$y = \frac{x}{e^T x + f}, \quad t = \frac{1}{e^T x + f}$$

The equivalent LP is given by:

$$\begin{array}{ll}\text{minimize} & c^T y + dt \\ \text{subject to} & Gy - ht \leq 0 \\ & Ay - bt = 0 \\ & e^T y + ft = 1 \\ & t \geq 0\end{array}$$

If (y, t) is the optimal solution for the LP and $t > 0$, then $x = y/t$ is the optimal solution for the linear-fractional problem.

Geometric Programming (GP).

Definition (Monomial). A function $f : \mathbb{R}_{++}^n \rightarrow \mathbb{R}$ defined as

$$f(x) = cx_1^{a_1} x_2^{a_2} \dots x_n^{a_n}$$

where $c > 0$ and $a_i \in \mathbb{R}$.

Definition (Posynomial). A sum of monomials:

$$f(x) = \sum_{k=1}^K c_k x_1^{a_{1k}} x_2^{a_{2k}} \dots x_n^{a_{nk}}$$

where $c_k > 0$.

Definition (Geometric Program (GP)). A geometric program is an optimization problem:

$$\begin{aligned} & \text{minimize} && f_0(x) \\ & \text{subject to} && f_i(x) \leq 1, \quad i = 1, \dots, m \\ & && h_i(x) = 1, \quad i = 1, \dots, p \end{aligned}$$

where $f_0, \dots, f_m$ are posynomials and $h_1, \dots, h_p$ are monomials. The domain is $\mathcal{D} = \mathbb{R}_{++}^n$.

GP is not convex in standard form, but can be transformed to a convex problem using a change of variables $y_i = \log x_i$ (so $x_i = e^{y_i}$) and taking the logarithm of the objective and constraints.

Definition (Convex form of GP). The transformed GP to a Convex GP is:

$$\begin{aligned} & \text{minimize} && \log f_0(e^y) \\ & \text{subject to} && \log f_i(e^y) \leq 0, \quad i = 1, \dots, m \\ & && \log h_i(e^y) = 0, \quad i = 1, \dots, p \end{aligned}$$

Since $\log(\sum e^{a_k^T y + b_k})$ is convex (Log-Sum-Exp function), this is a convex optimization problem.

Example. We have a following program:

$$\begin{aligned} & \text{minimize} && xy^{1/2} \\ & \text{subject to} && \frac{1}{3}x \leq 1 \\ & && 2x^{-1} \leq 1 \\ & && x^2 + 2yz^{-1} \leq 1 \\ & && xz^4 = 1 \\ & && x, y, z > 0 \end{aligned}$$

Here, the objective and inequality constraints involve posynomials (or monomials), and the equality constraint is a monomial.

- *Step 1: Change of Variables:* We introduce new variables x', y', z' such that:

$$x = e^{x'}, \quad y = e^{y'}, \quad z = e^{z'}$$

Taking the logarithm, we have:

$$x' = \log x, \quad y' = \log y, \quad z' = \log z$$

Note that the condition $x, y, z > 0$ is automatically satisfied since $e^u > 0$ for all real u .

- *Step 2: Transformation of Objective Function:* We substitute the variables into the objective function $f_0(x, y, z) = xy^{1/2}$:

$$f_0(e^{x'}, e^{y'}, e^{z'}) = e^{x'}(e^{y'})^{1/2} = e^{x'}e^{\frac{1}{2}y'} = e^{x' + \frac{1}{2}y'} \rightsquigarrow \text{minimize } e^{x' + \frac{1}{2}y'}$$

- *Step 3: Transformation of Constraints:* We transform each constraint by substituting variables and taking the logarithm (to achieve convexity).

$$\frac{1}{3}x \leq 1 : \quad \frac{1}{3}e^{x'} \leq 1 \iff \log(1/3) + x' \leq 0 \iff x' \leq \log 3 \quad (1)$$

$$2x^{-1} \leq 1 : \quad 2e^{-x'} \leq 1 \iff \log 2 - x' \leq 0 \iff x' \geq \log 2 \quad (2)$$

$$x^2 + 2yz^{-1} \leq 1 : \quad e^{2x'} + 2e^{y'-z'} \leq 1 \xrightarrow{\log} \log(e^{2x'} + e^{\log 2 + y' - z'}) \leq 0 \quad (3)$$

$$xz^4 = 1 : \quad e^{x'}(e^{z'})^4 = 1 \iff e^{x' + 4z'} = 1 \xrightarrow{\log} x' + 4z' = 0 \quad (4)$$

The equation (3) is in the form of the *Log-Sum-Exp* function (LSE), which is convex:

$$\text{lse}(2x', \log 2 + y' - z') \leq 0.$$

- *Step 4: Final Convex Problem in x', y', z' :* This is now a convex optimization problem:

$$\begin{aligned} &\text{minimize} && e^{x' + \frac{1}{2}y'} \quad (\text{or equivalently minimize } x' + \frac{1}{2}y') \\ &\text{subject to} && x' - \log 3 \leq 0 \\ &&& \log 2 - x' \leq 0 \\ &&& \log(e^{2x'} + e^{\log 2 + y' - z'}) \leq 0 \\ &&& x' + 4z' = 0 \end{aligned}$$

Observation. The bounds on x' immediately imply $\log 2 \leq x' \leq \log 3$, which simplifies the search space.

Theorem (Perron-Frobenius). Let $A \in \mathbb{R}^{n \times n}$ be a matrix with strictly positive entries ($A_{ij} > 0$).

- (1) There is a real eigenvalue $\lambda_{pf} > 0$ such that $\lambda_{pf} > |\lambda|$ for all other eigenvalues λ .
- (2) There exists a unique strictly positive eigenvector $v \succ 0$ (component-wise) such that $Av = \lambda_{pf}v$.

The condition that the growth rate is less than λ ($\rho(A) \leq \lambda$) is equivalent to the existence of a vector $v \succ 0$ such that:

$$\sum_{j=1}^n A_{ij}(u)v_j \leq \lambda v_i, \quad i = 1, \dots, n$$

This set of inequalities (vars u and v) involves posynomials and can be solved via GP to minimize λ .

Example (Bacteria Population). Consider a bacteria population $p_t \in \mathbb{R}_+^n$ at time t . The dynamics are given by:

$$p_{t+1} = A(u)p_t$$

where $A(u)$ is a matrix depending on chemical concentrations (resources) u . The entries $A_{ij}(u)$ are usually posynomials in u . The asymptotic growth rate is given by the spectral radius $\rho(A(u))$ (the largest eigenvalue modulus). We want to choose u to minimize (or target) the growth rate.

Quasiconvex Optimization.

Definition (Quasiconvex function). A function $f_0 : \mathbb{R}^n \rightarrow \mathbb{R}$ is quasiconvex if its sublevel sets

$$S_\alpha = \{x \in \text{dom } f_0 \mid f_0(x) \leq \alpha\}$$

are convex for all $\alpha \in \mathbb{R}$.

Example. Some typical examples:

- $\log(x)$ on $\mathbb{R}_{++}$ (concave, but also quasiconvex since sublevel sets are intervals $(0, e^\alpha]$).
- Ceiling function $\lceil x \rceil$: Quasiconvex (sublevel sets are intervals $(-\infty, k]$).
- Linear-fractional function: $f(x) = \frac{a^T x + b}{c^T x + d}$ on $\{x \mid c^T x + d > 0\}$.

Definition (Quasiconvex Optimization Problem).

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & Ax = b \end{array}$$

where f_0 is quasiconvex and $f_1, \dots, f_m$ are convex.

Observation. The feasible set X is convex. Solving the problem is equivalent to checking if the optimal value $p^* \leq \alpha$.

$$\exists x \in X, f_0(x) \leq \alpha \iff X \cap S_\alpha \neq \emptyset$$

Since S_α is convex (by definition of quasiconvexity) and X is convex, testing if $f_0(x) \leq \alpha$ is a convex feasibility problem.

Bisection Method.

Checking feasibility $\phi_\alpha(x) \leq 0$ is a convex problem, so we can find the optimal p^* by bisection on α .

Algorithm (Bisection method).

Input: interval $[l, u]$ containing p^* , and tolerance $\epsilon > 0$:

1. $\alpha := (l + u)/2$.
2. Solve convex feasibility problem: Find x s.t. $f_i(x) \leq 0, Ax = b, f_0(x) \leq \alpha$.
3. If feasible, set $u := \alpha$ (optimum is in lower half).
4. If infeasible, set $l := \alpha$ (optimum is in upper half).
5. Repeat until $u - l \leq \epsilon$.

Requires exactly $\lceil \log_2((u - l)/\epsilon) \rceil$ iterations.

Second-Order Cone Programming (SOCP).

SOCPs generalize Linear Programming (LP) and QCQP.

Definition (Second-Order Cone Program (SOCP)). A Second-Order Cone Program (SOCP) is an optimization problem of the form:

$$\begin{aligned} & \text{minimize} && f^T x \\ & \text{subject to} && \|A_i x + b_i\|_2 \leq c_i^T x + d_i, \quad i = 1, \dots, m \\ & && Fx = g \end{aligned}$$

where $x \in \mathbb{R}^n$. The constraints require the affine function $(A_i x + b_i, c_i^T x + d_i)$ to lie in the second-order cone (ice-cream cone) $\mathcal{K} = \{(y, t) \mid \|y\|_2 \leq t\}$.

Semidefinite Programming (SDP).

Definition (Semidefinite Programming (SDP)). SDP in Standard Form (Equational Form) is

$$\begin{aligned} & \text{minimize} && \text{tr}(CX) \\ & \text{subject to} && \text{tr}(A_i X) = b_i, \quad i = 1, \dots, p \\ & && X \succeq 0 \end{aligned}$$

where variable $X \in \mathbf{S}^n$ (symmetric matrix), and $C, A_1, \dots, A_p \in \mathbf{S}^n$. The constraint $X \succeq 0$ denotes that X is positive semidefinite (a generalized inequality with respect to the PSD cone).

The objective function uses the standard inner product on $\mathbf{S}^n$:

$$\langle C, X \rangle = \text{tr}(CX) = \sum_{i=1}^n \sum_{j=1}^n C_{ij} X_{ij}.$$

And the matrix $C \cdot X$ can be expressed as:

$$CX = \begin{pmatrix} \sum_{j=1}^n A_{1j} X_{j1} & & \\ & \ddots & \\ & & \sum_{j=1}^n A_{nj} X_{jn} \end{pmatrix}$$

Lemma. Let $X \in \mathbf{S}^n$. The following are equivalent:

- (1) $X \succeq 0$ (Positive Semidefinite)
- (2) $X = V^T V$ for some $V \in \mathbb{R}^{k \times n}$ (Gram matrix of columns of V)
- (3) $X = UU^T$ for some $U \in \mathbb{R}^{n \times k}$ (Sum of rank-1 matrices)

Proof. We prove implications (1) $\implies$ (3), (2) $\implies$ (1) and (3) $\implies$ (1):

- (1) $\implies$ (3): Since X is symmetric, it has a spectral decomposition $X = Q\Lambda Q^T$ with $\Lambda = \text{diag}(\lambda_i)$. If $X \succeq 0$, then $\lambda_i \geq 0$. Define $U = Q\Lambda^{1/2}$. Then $UU^T = Q\Lambda^{1/2}\Lambda^{1/2}Q^T = Q\Lambda Q^T = X$.
- (3) $\implies$ (2): Trivially.
- (2) $\implies$ (1): Let $X = UU^T$. For any $z \in \mathbb{R}^n$:

$$z^T X z = z^T U U^T z = (U^T z)^T (U^T z) = \|U^T z\|_2^2 \geq 0$$

Thus $X \succeq 0$.

Equivalence (2) $\Leftrightarrow$ (3) holds simply by taking $V = U^T$. □

Max-Cut problem.

Problem (Max-Cut). Given an undirected graph $G = (V, E)$ with weights $w_{ij} > 0$. Find a partition of vertices into two sets to maximize the sum of weights of edges crossing the cut.

For that, we define a *Quadratic Integer Formulation*:

Assign $x_i \in \{-1, 1\}$ to each vertex. Edge (i, j) is cut $\iff x_i \neq x_j$ (i.e., $x_i x_j = -1$).

$$\begin{aligned} & \text{maximize} && \frac{1}{2} \sum_{(i,j) \in E} w_{ij} (1 - x_i x_j) \\ & \text{subject to} && x_i \in \{-1, 1\}, \quad i = 1, \dots, n \end{aligned}$$

Since $x_i \in \{-1, 1\} \iff x_i^2 = 1$, we can rewrite this using a matrix $X = xx^T$. Note that $X_{ij} = x_i x_j$ and $X \succeq 0$, $\text{rank}(X) = 1$, $X_{ii} = 1$.

Now we provide *SDP-relaxation* of a Max-Cut problem.

Relax the hard constraint $\text{rank}(X) = 1$ (which makes the problem NP-hard).

$$\begin{aligned} & \text{maximize} && \frac{1}{2} \sum_{(i,j) \in E} w_{ij} (1 - X_{ij}) \\ & \text{subject to} && X_{ii} = 1, \quad i = 1, \dots, n \\ & && X \succeq 0 \end{aligned}$$

Using the first Lemma, $X_{ij} = v_i^T v_j$ where $v_i \in \mathbb{R}^n$ are unit vectors on the sphere ($\|v_i\|_2 = 1$). Geometric interpretation: We relax scalars $x_i \in \{-1, 1\}$ to vectors v_i on the unit sphere.

Lemma (Rank). $X \in \mathbf{S}_+^n \iff X = V^T V$ for some $V \in \mathbb{R}^{m \times n}$ where $m \leq n$.

Usually, we equate $X_{ij} = v_i^T v_j$, where v_i are the columns of V (vectors in $\mathbb{R}^m$).

Rounding Procedure.

How to convert the SDP solution X^* back to a valid cut x :

1. Compute decomposition $X^* = V^T V$, obtaining vectors $v_1, \dots, v_n$ on the unit sphere.
2. Choose a random hyperplane passing through the origin defined by normal vector r (uniformly distributed on sphere).
3. Assign $x_i = \mathbf{sign}(r^T v_i)$. (Vertices on one side of the plane are +1, others -1).

Proposition (Approximation Bounds). Let P^* be the true Max-Cut optimum, P_{SDP}^* the relaxation optimum, and P_{round} the expected value of the rounded solution.

$$P_{SDP}^* \geq P^* \geq P_{round} \geq 0.878 \cdot P_{SDP}^*$$

The SDP relaxation provides an upper bound, and the randomized rounding guarantees an approximation ratio of approx 0.878.

Vector Optimization

Definition (Vector Optimization Problem). *Minimizing a vector-valued objective function $f_0 : \mathbb{R}^n \rightarrow \mathbb{R}^q$ with respect to a proper cone $K \subseteq \mathbb{R}^q$ (typically $K = \mathbb{R}_+^q$).*

$$\begin{aligned} & \text{minimize (w.r.t. } K) && f_0(x) \\ & \text{subject to} && f_i(x) \leq 0, \quad i = 1, \dots, m \\ & && h_i(x) = 0, \quad i = 1, \dots, p \end{aligned}$$

Definition (Optimal Feasible Set (Pareto Optimal Points)). *A feasible point x is Pareto optimal if $f_0(x)$ is a minimal element of the set of achievable values – there is no feasible y such that $f_0(y) \preceq_K f_0(x)$ and $f_0(y) \neq f_0(x)$.*

In component-wise terms (for $K = \mathbb{R}_+^q$) x is Pareto optimal if it is not possible to decrease one objective $F_i(x)$ without increasing another $F_j(x)$.

Example. *Consider a bi-criterion problem:*

$$\text{minimize (w.r.t. } \mathbb{R}_+^2) \quad (\|Ax - b\|_2^2, \|x\|_2^2)$$

The two objectives compete: minimizing the residual (fitting data) vs. minimizing the norm (regularization).

Scalarization.

A standard technique to find Pareto optimal points is *scalarization*. We choose a weight vector $\lambda \succ_{K^*} 0$ (typically $\lambda > 0$ component-wise) and solve the standard convex problem:

$$\text{minimize } \lambda^T f_0(x) = \sum_{i=1}^q \lambda_i (f_0(x))_i$$

subject to original constraints.

Example.

$$\text{minimize } \lambda_1 \|Ax - b\|_2^2 + \lambda_2 \|x\|_2^2$$

Setting $\gamma = \lambda_2/\lambda_1$, this is Tikhonov regularization: $\min \|Ax - b\|_2^2 + \gamma \|x\|_2^2$.

Theorem. *If x is optimal for the scalarized problem with weights $\lambda \succ 0$, then x is Pareto optimal for the vector optimization problem.*

Proof. By contradiction. Assume x minimizes $\lambda^T f_0(z)$ over feasible z , but x is not Pareto optimal. Then there exists a feasible y such that $f_0(y) \preceq f_0(x)$ and $f_0(y) \neq f_0(x)$. Since $\lambda \succ 0$ and the difference $f_0(x) - f_0(y)$ has non-negative components with at least one positive component:

$$\lambda^T (f_0(x) - f_0(y)) > 0 \implies \lambda^T f_0(y) < \lambda^T f_0(x)$$

This contradicts the assumption that x minimizes the scalarized objective. □

Example (Portfolio Management). *Maximize return, minimize risk.*

- *Variables:* $x \in \mathbb{R}^n$ (portfolio weights).
- *Objectives:* $f_1(x) = -\bar{p}^T x$ (negative expected return), $f_2(x) = x^T \Sigma x$ (variance/risk).
- *Scalarization:* Minimize $x^T \Sigma x - \mu \bar{p}^T x$, where $\mu > 0$ is the risk aversion parameter.

Duality.

Consider an optimization problem in standard form (Primal P):

$$\begin{array}{ll}\text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & h_i(x) = 0, \quad i = 1, \dots, p\end{array}$$

Definition (Lagrangian). *The Lagrangian $L : \mathbb{R}^n \times \mathbb{R}^m \times \mathbb{R}^p \rightarrow \mathbb{R}$ is defined as:*

$$L(x, \lambda, \nu) = f_0(x) + \sum_{i=1}^m \lambda_i f_i(x) + \sum_{i=1}^p \nu_i h_i(x)$$

where $\lambda_i \geq 0$ are Lagrange multipliers (dual variables).

Definition (Lagrange dual function). *The Lagrange dual function $g : \mathbb{R}^m \times \mathbb{R}^p \rightarrow \mathbb{R}$ is the infimum of the Lagrangian over x :*

$$g(\lambda, \nu) = \inf_{x \in \mathcal{D}} L(x, \lambda, \nu).$$

Observation (Lower Bound Property). *If $\lambda \geq 0$ and x is feasible for the primal problem, then:*

$$L(x, \lambda, \nu) = f_0(x) + \sum \lambda_i f_i(x) + \sum \nu_i h_i(x) \leq f_0(x).$$

Proof. Since x is feasible, $f_i(x) \leq 0$ (so $\lambda_i f_i(x) \leq 0$) and $h_i(x) = 0$. Consequently, minimizing over all x gives:

$$g(\lambda, \nu) \leq L(x, \lambda, \nu) \leq f_0(x)$$

Therefore, $g(\lambda, \nu) \leq p^*$ (the optimal value of the primal problem). This is called *Weak Duality*. $\square$

Observation (Concavity). *The dual function $g(\lambda, \nu)$ is concave, even if the primal problem is not convex.*

It's because it is the pointwise infimum of a family of affine functions of (λ, ν) .

The Dual Problem.

Definition (Lagrange Dual Problem). *The Lagrange Dual Problem is to find the best lower bound:*

$$\begin{array}{ll}\text{maximize} & g(\lambda, \nu) \\ \text{subject to} & \lambda \succeq 0\end{array}$$

Example (Least Squares with Equality Constraints).

$$\begin{array}{ll}\text{minimize} & x^T x \\ \text{subject to} & Ax = b\end{array}$$

We can solve it by the following steps:

- Lagrangian: $L(x, \nu) = x^T x + \nu^T (Ax - b)$.
- Minimize L w.r.t x : $\nabla_x L = 2x + A^T \nu = 0 \implies x = -(1/2)A^T \nu$.
- Dual Function $g(\nu)$: Substitute x back into L :

$$g(\nu) = \frac{1}{4} \nu^T A A^T \nu + \nu^T (A(-(1/2)A^T \nu) - b) = -\frac{1}{4} \nu^T A A^T \nu - \nu^T b$$

- Dual Problem: Maximize $-\frac{1}{4} \nu^T A A^T \nu - \nu^T b$ (unconstrained concave quadratic).

Example (LP in Standard Form).

$$\begin{aligned} & \text{minimize} && c^T x \\ & \text{subject to} && Ax = b \\ & && -x \leq 0 \end{aligned}$$

We can solve it by the following steps:

- Lagrangian: $L(x, \lambda, \nu) = c^T x + \nu^T (Ax - b) + \lambda^T (-x) = (c + A^T \nu - \lambda)^T x - b^T \nu$.
- Dual Function:

$$g(\lambda, \nu) = \inf_x ((c + A^T \nu - \lambda)^T x - b^T \nu)$$

This is linear in x . If the coefficient vector $(c + A^T \nu - \lambda) \neq 0$, the infimum is $-\infty$. If it is 0, the infimum is $-b^T \nu$.

$$g(\lambda, \nu) = \begin{cases} -b^T \nu & \text{if } A^T \nu - \lambda + c = 0 \\ -\infty & \text{otherwise} \end{cases}$$

- Dual Problem is:

$$\begin{aligned} & \text{maximize} && -b^T \nu \\ & \text{subject to} && A^T \nu + c \geq 0. \end{aligned}$$

Strong Duality.

Example (Quadratic problem). *Consider the problem:*

$$\begin{aligned} & \text{minimize} && x^2 + 1 \\ & \text{subject to} && (x - 2)(x - 4) \leq 0 \end{aligned}$$

The feasible set is $[2, 4]$. The optimal value is $p^* = 5$ (attained at $x = 2$).

The Lagrangian is $L(x, \lambda) = x^2 + 1 + \lambda(x^2 - 6x + 8)$. Minimizing w.r.t. x (unconstrained quadratic):

$$\nabla_x L = 2x + 2\lambda x - 6\lambda = 0 \implies x = \frac{3\lambda}{1 + \lambda}$$

Substituting back, we get the dual function:

$$g(\lambda) = -9 \frac{\lambda^2}{1 + \lambda} + 1 + 8\lambda$$

Maximizing $g(\lambda)$ for $\lambda \geq 0$ yields $\lambda^* = 2$, and dual optimum $d^* = g(2) = 5$. Here, $p^* = d^*$ (Strong Duality holds).

Definition (Weak and Strong duality). *Let p^* be the optimal value of the primal problem and d^* be the optimal value of the dual problem.*

- Weak Duality: $d^* \leq p^*$ (Always holds).
- Strong Duality: $d^* = p^*$.

Strong duality does not hold in general, but usually holds for convex problems under constraint qualifications.

Slater's Condition.

Definition (Strict feasibility). *A problem is strictly feasible if $\exists x \in \text{int}(\mathcal{D})$ such that:*

$$f_i(x) < 0, \quad i = 1, \dots, m \quad (\text{for non-affine constraints})$$

Theorem (Slater's). *If the primal problem is convex and strictly feasible, then strong duality $p^* = d^*$ holds.*

Observation (Strong duality for LP). *For Linear Programs, strong duality holds if the primal is feasible (or if the dual is feasible). Strict feasibility is not required, only feasibility.*

Geometric Interpretation

Define the set of values $G \subseteq \mathbb{R}^2$ (for one inequality constraint):

$$G = \{(f_1(x), f_0(x)) \mid x \in \mathcal{D}\}$$

And the set $\mathcal{A} = G + \mathbb{R}_+^2$ (epigraph variation):

$$\mathcal{A} = \{(u, t) \mid \exists x \in \mathcal{D}, f_1(x) \leq u, f_0(x) \leq t\}$$

The optimal value is $p^* = \inf\{t \mid (0, t) \in \mathcal{A}\}$.

Observation (Dual function geometry). *The dual function at $\lambda \geq 0$ can be interpreted using supporting hyperplanes:*

$$g(\lambda) = \inf_{(u,t) \in G} (t + \lambda u)$$

The inequality $t + \lambda u \geq g(\lambda)$ defines a supporting hyperplane to $\mathcal{A}$ with normal vector $(\lambda, 1)$.

Theorem (Strong Duality condition). *Strong duality holds $\iff$ there exists a non-vertical supporting hyperplane to $\mathcal{A}$ at the point $(0, p^*)$.*

If the problem is convex, $\mathcal{A}$ is a convex set. If a convex set has a boundary point $(0, p^*)$, there exists a supporting hyperplane. Slater's condition ensures this hyperplane is not vertical.

Saddle Point Interpretation.

Theorem (Max-Min inequality). *For any function $f : W \times Z \rightarrow \mathbb{R}$:*

$$\sup_{z \in Z} \inf_{w \in W} f(w, z) \leq \inf_{w \in W} \sup_{z \in Z} f(w, z).$$

Proof Idea. Let $g(z) = \inf_w f(w, z)$. Then $g(z) \leq f(w, z)$ for all w, z . Taking sup over z , $\sup g(z) \leq \sup_z f(w, z)$. Then taking inf over w , $\sup_z \inf_w f \leq \inf_w \sup_z f$. $\square$

We can see that since the *Primal optimum* is $p^* = \inf_x \sup_{\lambda \geq 0} L(x, \lambda)$ and *Dual optimum* $d^* = \sup_{\lambda \geq 0} \inf_x L(x, \lambda)$, by the max-min inequality $d^* \leq p^*$. Strong duality means equality holds.

Definition (Saddle point). *A pair (x^*, λ^*) is a saddle point of the Lagrangian L if $\forall x, \lambda \succeq 0$:*

$$L(x^*, \lambda) \leq L(x^*, \lambda^*) \leq L(x, \lambda^*)$$

This implies x^ minimizes $L(x, \lambda^*)$ and λ^* maximizes $L(x^*, \lambda)$.*

Complementary Slackness.

Assume that strong duality holds, let x^* be primal optimal and λ^* be dual optimal.

$$f_0(x^*) = g(\lambda^*) = \inf_x (f_0(x) + \sum \lambda_i^* f_i(x)) \leq f_0(x^*) + \sum \lambda_i^* f_i(x^*) \leq f_0(x^*)$$

The first inequality comes from the definition of infimum, the second from feasibility ($f_i(x^*) \leq 0, \lambda_i^* \geq 0$). For equality to hold, we must have:

$$\sum_{i=1}^m \lambda_i^* f_i(x^*) = 0$$

Since each term is non-positive, each term must be zero:

$$\lambda_i^* f_i(x^*) = 0, \quad i = 1, \dots, m$$

This is the **complementary slackness** condition. It means either $\lambda_i^* = 0$ (constraint inactive) or $f_i(x^*) = 0$ (constraint active).

KKT Conditions.

For a convex optimization problem (or generally any optimization problem with differentiable objective and constraint functions), the KKT conditions for points $x \in \mathbb{R}^n, \lambda \in \mathbb{R}^m, \nu \in \mathbb{R}^p$ are:

1. **Primal Feasibility:**

$$\begin{aligned} f_i(x) &\leq 0, \quad i = 1, \dots, m \\ h_i(x) &= 0, \quad i = 1, \dots, p \end{aligned}$$

2. **Dual Feasibility:**

$$\lambda_i \geq 0, \quad i = 1, \dots, m$$

3. **Complementary Slackness:**

$$\lambda_i f_i(x) = 0, \quad i = 1, \dots, m$$

4. **Stationarity (Gradient of Lagrangian vanishes):**

$$\nabla f_0(x) + \sum_{i=1}^m \lambda_i \nabla f_i(x) + \sum_{i=1}^p \nu_i \nabla h_i(x) = 0$$

Theorem (Necessity of KKT Conditions). *If strong duality holds and x^* is primal optimal and (λ^*, ν^*) are dual optimal, then the pair $(x^*, (\lambda^*, \nu^*))$ must satisfy the KKT conditions.*

Proof idea. It follows from the minimization of the Lagrangian $L(x, \lambda^*, \nu^*)$ at x^* (implies stationarity) and the zero duality gap (implies complementary slackness). $\square$

Theorem (Sufficiency of KKT Conditions). *If the optimization problem is convex (objective and inequality constraints are convex, equality constraints affine), then any points $\tilde{x}, \tilde{\lambda}, \tilde{\nu}$ that satisfy the KKT conditions are primal and dual optimal, and strong duality holds (zero duality gap).*

For convex problems with differentiable functions that satisfy Slater's condition, the KKT conditions are necessary and sufficient for optimality.

Farkas' Lemma.

Lemma (Farkas'). *Let $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$. Then exactly one of the following two systems has a solution:*

- (1) $\exists x \in \mathbb{R}^n$ such that $Ax = b$ and $x \geq 0$.
- (2) $\exists y \in \mathbb{R}^m$ such that $A^T y \geq 0$ and $y^T b < 0$.

Proof. Suppose both systems have solutions x and y . Consider the quantity $y^T Ax$. We can evaluate it in two ways:

1. Using system (1): $y^T(Ax) = y^T b < 0$ (since $b^T y < 0$ from system 2).
2. Using system (2): $(y^T A)x = (A^T y)^T x$. Since $A^T y \geq 0$ and $x \geq 0$, the inner product of two non-negative vectors is non-negative: $(A^T y)^T x \geq 0$.

By that, we obtain $0 \leq (A^T y)^T x = y^T b < 0$, which is impossible. Thus, both cannot hold simultaneously.

Now assume that System (1) has no solution. We must show System (2) has a solution. Define the set C as the cone generated by the columns of A :

$$C = \{Ax \mid x \geq 0\} = \left\{ \sum_{i=1}^n x_i a_i \mid x_i \geq 0 \right\}$$

C is a non-empty closed convex cone. Since System (1) has no solution, vector b does not belong to C ($b \notin C$).

By the *Separating Hyperplane Theorem* (since C is closed and convex and $b \notin C$), there exists a hyperplane that strictly separates b from C . This means there exists a vector $y \in \mathbb{R}^m$ and a scalar α such that:

$$y^T b < \alpha \leq y^T z \quad \forall z \in C$$

We can see that since $0 \in C$ (choose $x = 0$), we have $\alpha \leq y^T 0 = 0$. Thus, $y^T b < 0$. This satisfies the second part of System (2).

And we need to show $A^T y \geq 0$. Suppose for contradiction that for some column a_j of A , $y^T a_j < 0$. Since $a_j \in C$, any positive multiple λa_j is also in C (cone property). Let $z = \lambda a_j$ for arbitrarily large $\lambda > 0$. Then $y^T z = \lambda(y^T a_j) \rightarrow -\infty$ as $\lambda \rightarrow \infty$. This contradicts the lower bound $y^T z \geq \alpha$. Therefore, $y^T a_j \geq 0$ for all columns $j = 1, \dots, n$. In matrix form: $A^T y \geq 0$. $\square$

Geometric interpretation of the two systems above is the following:

- System (1) says that b lies in the *cone* generated by the columns of A (denoted $\text{cone}(A)$).
- System (2) says that there is a hyperplane (defined by normal y) that separates b from $\text{cone}(A)$. Specifically, the projection of columns of A onto y is non-negative (y makes an acute angle with them), but the projection of b onto y is negative (y makes an obtuse angle with b).

Theorem (Strong Duality). *There are four possibilities:*

- (1) *Both primal and dual have no feasible solutions (are infeasible).*
- (2) *The primal is infeasible and the dual unbounded.*
- (3) *The dual is infeasible and the primal unbounded.*
- (4) *Both primal and dual have feasible solutions and their values are equal.*

Proof. Assume we have two programs:

<i>Primal LP</i>	<i>Dual LP</i>
minimize $c^T x$ such that $Ax = b$ $x \geq 0$	maximize $b^T y$ such that $A^T y \leq c$

Assume Primal has an optimal solution x^* . We want to show there exists a dual feasible y^* such that $c^T x^* = b^T y^*$.

Suppose (for contradiction) there is no such dual solution that achieves the value $\gamma = c^T x^*$. This can be formulated as a system of inequalities. Using a variant of Farkas' Lemma on the system combining dual feasibility and the optimality requirement ($b^T y \geq \gamma$), we can derive a contradiction or show that if the system has no solution, the primal would typically be unbounded or infeasible, contradicting the assumption of optimality. Specifically, Farkas' Lemma is the key tool to rigorously prove that if a Linear Program is feasible and bounded, the duality gap is zero ($p^* = d^*$). $\square$

Sensitivity Analysis.

Definition (The Perturbed Problem). Consider the standard optimization problem (unperturbed) and its limits. We define the **perturbed problem** by relaxing the constraints limits:

$$\begin{aligned} & \text{minimize} && f_0(x) \\ & \text{subject to} && f_i(x) \leq u_i, \quad i = 1, \dots, m \\ & && h_i(x) = v_i, \quad i = 1, \dots, p \end{aligned}$$

where $u \in \mathbb{R}^m$ and $v \in \mathbb{R}^p$ are parameters. We define $p^*(u, v)$ as the optimal value of this perturbed problem. Note that $p^*(0, 0)$ is the optimal value of the original problem.

Global Inequality (Lower Bound).

Assume strong duality holds for the original problem (at $u = 0, v = 0$) and let (λ^*, ν^*) be the optimal dual variables. For any x feasible for the perturbed problem ($f_i(x) \leq u_i, h_i(x) = v_i$):

$$\begin{aligned} p^*(0, 0) = g(\lambda^*, \nu^*) &\leq f_0(x) + \sum_{i=1}^m \lambda_i^* f_i(x) + \sum_{i=1}^p \nu_i^* h_i(x) \\ &\leq f_0(x) + \sum_{i=1}^m \lambda_i^* u_i + \sum_{i=1}^p \nu_i^* v_i \end{aligned}$$

Result: Since this holds for any feasible x in the perturbed problem:

$$p^*(u, v) \geq p^*(0, 0) - \sum_{i=1}^m \lambda_i^* u_i - \sum_{i=1}^p \nu_i^* v_i$$

Interpretation: Tightening constraints ($u_i < 0$) worsens the optimal value (increases p^*) by at least $\lambda_i^* |u_i|$. Loosening ($u_i > 0$) improves it by at most $\lambda_i^* u_i$.

Local Analysis (Gradient).

If $p^*(u, v)$ is differentiable at $(0, 0)$, then:

$$\lambda_i^* = -\frac{\partial p^*(0, 0)}{\partial u_i}, \quad \nu_i^* = -\frac{\partial p^*(0, 0)}{\partial v_i}$$

The Lagrange multipliers λ_i^* act as *shadow prices*, representing the rate of change of the optimal value with respect to perturbations in constraints.

Approximation and Fitting.

We want to approximate a vector b by a linear combination of columns of A , i.e., $Ax \approx b$. Common contexts:

1. **Geometric approximation:** b is a target point, columns of A are directions.
2. **Optimal design:** x are design variables.
3. **Optimal estimation:** Extracting parameters x from noisy measurements b .

Least Squares Approximation.

$$\text{minimize} \quad \|Ax - b\|_2^2 = (Ax - b)^T(Ax - b)$$

Expanding the term:

$$x^T A^T A x - 2b^T A x + b^T b$$

This is a convex quadratic function. The optimal solution satisfies the condition (gradient = 0):

$$A^T A x = A^T b \quad (\text{Normal Equations})$$

Definition (Penalty Function Approximation). *We solve the problem:*

$$\text{minimize} \quad \sum_{i=1}^m \phi(r_i), \quad \text{where } r = Ax - b$$

Here, $\phi : \mathbb{R} \rightarrow \mathbb{R}$ is a penalty function. It is typically convex, $\phi(0) = 0$, $\phi(u) \geq 0$, and symmetric $\phi(u) = \phi(-u)$.

Example. *Some examples of penalty functions:*

- Quadratic (Least Squares): $\phi(u) = u^2$. *Heavily penalizes large errors.*
- Deadzone-linear:

$$\phi(u) = \max(0, |u| - a)$$

Ignores errors smaller than a . Used when small deviations are acceptable.

- Log-barrier: *(Used for constraints, or in context of approximation as $-\log(a^2 - u^2)$ type forms to enforce bounds, though user likely referred to barrier-like penalties).*

Least squares ($\phi(u) = u^2$) is sensitive to outliers because the penalty grows quadratically. To deal with outliers, we use penalty functions that grow linearly (or less than quadratically) for large u , such as the ℓ_1 -norm ($\phi(u) = |u|$).

Huber Penalty Function.

The Huber function combines the smoothness of Least Squares for small errors and the robustness of ℓ_1 for large errors (outliers):

$$\phi_{\text{hub}}(u) = \begin{cases} u^2 & |u| \leq M \\ M(2|u| - M) & |u| > M \end{cases}$$

This function is convex and differentiable everywhere.

Example (Signal Denoising (Reconstruction)). *We observe a noisy signal $y \in \mathbb{R}^n$, which is a corrupted version of the true signal x :*

$$y = x + v$$

where v is noise. We want to estimate x . Since n (variables) equals the number of measurements, we need prior information (regularization) to solve this.

We can formulize it as follows:

$$\text{minimize} \quad \|\hat{x} - y\|_2^2 + \gamma \phi(\hat{x})$$

- First term: Fidelity (keep estimate close to observation).
- Second term $\phi(\hat{x})$: Regularization (enforce structure, e.g., smoothness).
- $\gamma > 0$: Trade-off parameter.

Types of regularization:

- *Quadratic Smoothing* (ℓ_2): $\phi(\hat{x}) = \sum_{i=1}^{n-1} (\hat{x}_{i+1} - \hat{x}_i)^2 = \|D\hat{x}\|_2^2$. Result: Smooths the signal (like a low-pass filter), blurs sharp edges.
- *Total Variation Denoising* (ℓ_1): $\phi(\hat{x}) = \sum_{i=1}^{n-1} |\hat{x}_{i+1} - \hat{x}_i| = \|D\hat{x}\|_1$. Result: Preserves sharp transitions (edges) while removing noise. The solution tends to be piecewise constant.

Example (Robust Approximation). *The goal is to solve $Ax \approx b$ in the presence of outliers (measurements with arbitrarily large errors).*

Comparison of norms:

- *Least Squares* (ℓ_2): Minimize $\|Ax - b\|_2$. Sensitive to outliers because large residuals are squared. Ideally assumes Gaussian noise.
- *Chebyshev Approximation* (ℓ_∞): Minimize $\|Ax - b\|_\infty$. Extremely sensitive to outliers (determined entirely by the worst data point).
- *Sum of Absolute Residuals* (ℓ_1): Minimize $\|Ax - b\|_1$. *Robust*. The optimal solution is unchanged even if extreme outliers are moved further away, as long as their sign (relative to the fit) doesn't change.

LP Formulation of ℓ_1 -norm approximation:

$$\begin{aligned} & \text{minimize} && \mathbf{1}^T t \\ & \text{subject to} && -t \preceq Ax - b \preceq t \end{aligned}$$

where $t \in \mathbb{R}^m$ are auxiliary variables.

Example (Optimal Input Design). *We want to estimate a vector x from measurements $y_i = a_i^T x + v_i$. The error covariance of the Least Squares estimator is proportional to $(A^T A)^{-1}$, where A has rows a_i^T . In experiment design, we choose the vectors a_i from a set of available test vectors $V = \{v_1, \dots, v_K\}$ to make the estimation as accurate as possible.*

Let m_k be the number of times we choose vector v_k .

$$A^T A = \sum_{k=1}^K m_k v_k v_k^T$$

Relaxing m_k to continuous variables $\lambda_k = m_k/m$ (frequency of use), we optimize a scalar measure of the "size" of the error covariance matrix $E = (\sum \lambda_k v_k v_k^T)^{-1}$.

Criteria:

- *D-optimal design*: Minimize $\log \det E$ (minimize the volume of the confidence ellipsoid).
- *E-optimal design*: Minimize $\lambda_{\max}(E)$ (minimize the diameter of the confidence ellipsoid).
- *A-optimal design*: Minimize $\text{tr}(E)$ (minimize the mean squared error).

This is a convex problem in $\lambda \in \mathbb{R}^K$ (subject to $\lambda \geq 0, \sum \lambda_k = 1$).

Example (Function Fitting and Sparse Regression). *We want to fit a function $f: \mathbb{R}^k \rightarrow \mathbb{R}$ to data points (u_i, y_i) by minimizing measurement error.*

$$f(u) = \sum_{j=1}^n x_j \psi_j(u)$$

where ψ_j are basis functions (e.g., polynomials, splines, trig functions). The problem reduces to finding coefficients x .

Sparse Fitting (Lasso): If we want a simple model (few basis functions used), we add ℓ_1 regularization:

$$\text{minimize} \quad \sum_{i=1}^m (f(u_i) - y_i)^2 + \lambda \|x\|_1$$

This forces many coefficients x_j to be exactly zero, effectively selecting the "best" subset of basis functions.

Maximum Likelihood (ML) Estimation.

Definition (Log-likelihood function). *We consider a family of probability distributions $p_x(y)$ on $\mathbb{R}^m$, indexed by a parameter $x \in \mathcal{X} \subseteq \mathbb{R}^n$. Given an observation $y \in \mathbb{R}^m$, the log-likelihood function is defined as:*

$$l(x) = \log p_x(y)$$

Definition (Maximum Likelihood (ML) estimate). *The Maximum Likelihood (ML) estimate is the parameter x that maximizes the probability of observing y :*

$$\hat{x}_{ml} = \arg \max_{x \in \mathcal{X}} l(x) = \arg \max_{x \in \mathcal{X}} \log p_x(y)$$

If the log-likelihood function $l(x)$ is concave (and the domain $\mathcal{X}$ is convex), then ML estimation is a convex optimization problem.

Linear Measurement Models.

Consider the linear model:

$$y_i = a_i^T x + v_i, \quad i = 1, \dots, m$$

where $x \in \mathbb{R}^n$ is the vector of parameters to be estimated, y_i are measurements, and v_i are independent, identically distributed (IID) noise terms with density $p(z)$.

The *likelihood* function is:

$$p_x(y) = \prod_{i=1}^m p(y_i - a_i^T x)$$

The *log-likelihood* is:

$$l(x) = \sum_{i=1}^m \log p(y_i - a_i^T x)$$

- **Gaussian Noise (Least Squares):** If the noise is Gaussian, $v_i \sim \mathcal{N}(0, \sigma^2)$, then

$$p(z) = (2\pi\sigma^2)^{-1/2} e^{-z^2/2\sigma^2}.$$

$$l(x) = -\frac{m}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^m (y_i - a_i^T x)^2$$

Maximizing $l(x)$ is equivalent to minimizing the sum of squared errors:

$$\text{minimize} \quad \|Ax - y\|_2^2$$

Thus, ML estimation with IID Gaussian noise leads to the *Least Squares* problem.

- **Laplacian Noise (ℓ_1 -norm):** If the noise has a Laplacian distribution $p(z) = (1/2a)e^{-|z|/a}$, maximizing $l(x)$ is equivalent to minimizing the ℓ_1 -norm of residuals:

$$\text{minimize} \quad \|Ax - y\|_1$$

This is robust to outliers.

Logistic Regression.

Used for binary classification. We observe pairs (u_i, y_i) where $u_i \in \mathbb{R}^n$ are features and $y_i \in \{0, 1\}$ are binary outcomes. We model the probability of $y = 1$ as:

$$p_1(u) = \Pr(y = 1 \mid u) = \frac{\exp(a^T u + b)}{1 + \exp(a^T u + b)}$$

$$p_0(u) = \Pr(y = 0 \mid u) = 1 - p_1(u) = \frac{1}{1 + \exp(a^T u + b)}$$

Parameters to estimate are $x = (a, b)$.

Optimization Problem: The log-likelihood for N observations is:

$$l(a, b) = \sum_{i=1}^N (y_i \log p_1(u_i) + (1 - y_i) \log p_0(u_i))$$

Substituting the logistic function:

$$l(a, b) = \sum_{i=1}^N (y_i(a^T u_i + b) - \log(1 + \exp(a^T u_i + b)))$$

This function is *concave* in (a, b) , so finding the ML estimate is a convex optimization problem.

Covariance Matrix Estimation.

We observe samples $y_1, \dots, y_N \in \mathbb{R}^n$ drawn from a Gaussian distribution $\mathcal{N}(0, \Sigma)$. We want to estimate the covariance matrix $\Sigma \in \mathbf{S}_{++}^n$.

The log-likelihood is:

$$l(\Sigma) = -\frac{Nn}{2} \log(2\pi) - \frac{N}{2} \log \det \Sigma - \frac{1}{2} \sum_{i=1}^N y_i^T \Sigma^{-1} y_i$$

This is not concave in Σ . However, we can reparametrize using the *inverse covariance matrix* (precision matrix) $S = \Sigma^{-1}$.

$$l(S) = \frac{N}{2} \log \det S - \frac{1}{2} \sum_{i=1}^N \text{tr}(S y_i y_i^T) + \text{const}$$

$$l(S) = \frac{N}{2} (\log \det S - \text{tr}(SY)) + \text{const}$$

where $Y = \frac{1}{N} \sum y_i y_i^T$ is the sample covariance matrix. Since $\log \det S$ is concave on $\mathbf{S}_{++}^n$, this is a *convex optimization problem* in variable S .

Binary Hypothesis Testing.

Problem Setup.

We want to decide between two hypotheses about a random variable x :

- H_0 : x is generated from probability distribution $p_0(x)$.
- H_1 : x is generated from probability distribution $p_1(x)$.

We design a decision rule (detector) or a randomized detector. For a discrete variable $x \in \{1, \dots, n\}$, a randomized detector is defined by a matrix $T \in \mathbb{R}^{2 \times n}$, where T_{ik} is the probability of choosing hypothesis H_i given observing event k .

Error Probabilities

- **False Positive (Type I Error):** Deciding H_1 when H_0 is true.

$$P_{fp} = \sum_x \Pr(\text{choose } H_1 \mid x) p_0(x)$$

- **False Negative (Type II Error):** Deciding H_0 when H_1 is true.

$$P_{fn} = \sum_x \Pr(\text{choose } H_0 \mid x) p_1(x)$$

Neyman-Pearson Lemma (Convex Formulation).

We want to minimize the False Negative rate subject to a limit α on the False Positive rate.

$$\begin{array}{ll} \text{minimize} & P_{fn} \\ \text{subject to} & P_{fp} \leq \alpha \end{array}$$

For a discrete distribution, this is a *Linear Program* in variables $t_k = \Pr(\text{choose } H_1 \mid x = k)$ (where $0 \leq t_k \leq 1$):

$$\begin{array}{ll} \text{minimize} & \sum_{k=1}^n (1 - t_k) p_1(k) \\ \text{subject to} & \sum_{k=1}^n t_k p_0(k) \leq \alpha \\ & 0 \leq t_k \leq 1, \quad k = 1, \dots, n \end{array}$$

Solution: The optimal detector is a deterministic **Likelihood Ratio Test** (possibly randomized at the threshold):

$$\frac{p_1(x)}{p_0(x)} \geq \lambda \implies \text{choose } H_1$$

Geometric Problems: Löwner-John Ellipsoids

Maximum Volume Ellipsoid in a Polyhedron

We want to find the largest ellipsoid $\mathcal{E}$ contained in a polyhedron $P = \{x \mid a_i^T x \leq b_i, i = 1, \dots, m\}$.

Parametrization: We represent the ellipsoid as the image of the unit ball under an affine mapping:

$$\mathcal{E} = \{Bu + d \mid \|u\|_2 \leq 1\}$$

where $B \in \mathbf{S}_{++}^n$ (symmetric positive definite) and $d \in \mathbb{R}^n$ (center). The volume of $\mathcal{E}$ is proportional to $\det B$.

Constraints: The condition $\mathcal{E} \subseteq P$ means that for all u with $\|u\|_2 \leq 1$:

$$a_i^T(Bu + d) \leq b_i \iff \sup_{\|u\|_2 \leq 1} (a_i^T Bu) + a_i^T d \leq b_i$$

Using the dual norm definition, $\sup_{\|u\| \leq 1} z^T u = \|z\|_2$:

$$\|Ba_i\|_2 + a_i^T d \leq b_i, \quad i = 1, \dots, m$$

Convex Optimization Problem: To maximize volume, we maximize $\log \det B$ (is concave).

$$\begin{aligned} & \text{maximize} && \log \det B \\ & \text{subject to} && \|Ba_i\|_2 + a_i^T d \leq b_i, \quad i = 1, \dots, m \end{aligned}$$

Minimum Volume Ellipsoid covering a Finite Set (Löwner Ellipsoid).

We want to find the smallest ellipsoid $\mathcal{E}$ containing a set of points $C = \{x_1, \dots, x_m\}$ (or equivalently their convex hull).

Parametrization: It is convenient to parameterize the ellipsoid as:

$$\mathcal{E} = \{x \mid \|Ax + b\|_2 \leq 1\}$$

where $A \in \mathbf{S}_{++}^n$. The volume is proportional to $\det A^{-1}$. To minimize volume, we maximize $\det A$ (or $\log \det A$).

Constraints: The condition $x_i \in \mathcal{E}$ is simply:

$$\|Ax_i + b\|_2 \leq 1, \quad i = 1, \dots, m$$

Convex Optimization Problem:

$$\begin{aligned} & \text{maximize} && \log \det A \\ & \text{subject to} && \|Ax_i + b\|_2 \leq 1, \quad i = 1, \dots, m \end{aligned}$$

Note: This allows computing the Löwner-John ellipsoid for a polyhedron described by its vertices.

Approximation of Convex Sets (Löwner-John Theorem)

Let $C \subset \mathbb{R}^n$ be a bounded convex set with non-empty interior. Let $\mathcal{E}_{inner}$ be the max volume ellipsoid in C (John ellipsoid). Let $\mathcal{E}_{outer}$ be the min volume ellipsoid covering C (Löwner ellipsoid).

Theorem.

- For the John ellipsoid (centered at x_0):

$$\mathcal{E}_{inner} \subseteq C \subseteq n(\mathcal{E}_{inner} - x_0) + x_0$$

(If C is symmetric about the center, the factor improves to $\sqrt{n}$).

- For the Löwner ellipsoid:

$$\frac{1}{n}(\mathcal{E}_{outer} - x_0) + x_0 \subseteq C \subseteq \mathcal{E}_{outer}$$

This provides a bound on how well any convex set can be approximated by an ellipsoid (efficiency of affine invariant algorithms).

Centers of Polyhedra.

We consider a polyhedron $P = \{x \mid a_i^T x \leq b_i, i = 1, \dots, m\}$. We want to find a center point $x_c \in P$.

1. **Center of Maximum Volume Ellipsoid.** We can define the center of P as the center d of the maximum volume ellipsoid $\mathcal{E} = \{Bu + d \mid \|u\|_2 \leq 1\}$ contained in P . This requires solving the convex problem discussed in the previous lecture (maximize $\log \det B$).
2. **Chebyshev Center.** It is the center of the largest Euclidean ball $\mathcal{B} = \{x_c + u \mid \|u\|_2 \leq r\}$ contained in P . We want to maximize the radius r . *LP Formulation:*

$$\begin{aligned} & \text{maximize} && r \\ & \text{subject to} && a_i^T x_c + r \|a_i\|_2 \leq b_i, \quad i = 1, \dots, m \\ & && r \geq 0 \end{aligned}$$

Variables are r and x_c . Since $\|a_i\|_2$ are constants, this is a Linear Program.

3. **Analytic Center.** It is the point that maximizes the product of distances to the defining hyperplanes (or minimizes the log-barrier function).

$$\text{minimize} \quad - \sum_{i=1}^m \log(b_i - a_i^T x)$$

Implicitly requires $x \in \text{int } P$. This is an unconstrained convex minimization problem.

Support Vector Machines (Classification).

Linear Discrimination.

Given two sets of points $X = \{x_1, \dots, x_N\}$ and $Y = \{y_1, \dots, y_M\}$ in $\mathbb{R}^n$. We want to find an affine function $f(z) = a^T z - b$ (hyperplane) that separates them:

$$f(x_i) > 0 \quad \forall i, \quad f(y_j) < 0 \quad \forall j$$

This is equivalent to finding a, b such that:

$$a^T x_i - b \geq 1, \quad a^T y_j - b \leq -1$$

Robust Linear Discrimination (Max Margin).

If the sets are separable, there are many separating hyperplanes. We want the one that maximizes the **margin** (distance to the closest point). We have two parallel hyperplanes $H_1 = \{z \mid a^T z - b = 1\}$ and $H_2 = \{z \mid a^T z - b = -1\}$, the distance $\text{dist}(H_1, H_2) = 2/\|a\|_2$. Maximizing margin $\iff$ Minimizing $\|a\|_2$.

Optimization Problem (Hard Margin SVM): – Quadratic Program

$$\begin{aligned} & \text{minimize} && \|a\|_2 \\ & \text{subject to} && a^T x_i - b \geq 1, \quad i = 1, \dots, N \\ & && a^T y_j - b \leq -1, \quad j = 1, \dots, M \end{aligned}$$

Approximate Linear Discrimination (Soft Margin).

If the sets are not linearly separable, we allow some violations using slack variables $u_i \geq 0$ and $v_j \geq 0$.

$$a^T x_i - b \geq 1 - u_i, \quad a^T y_j - b \leq -(1 - v_j)$$

We minimize the sum of violations (and typically the norm $\|a\|_2$ for regularization).

LP Formulation (Minimize total violation):

$$\begin{aligned} & \text{minimize} && \sum_{i=1}^N u_i + \sum_{j=1}^M v_j \\ & \text{subject to} && a^T x_i - b \geq 1 - u_i \\ & && a^T y_j - b \leq -1 + v_j \\ & && u_i \geq 0, v_j \geq 0 \end{aligned}$$

Non-linear Discrimination (Polynomials).

We can separate points using a non-linear function $f(z)$ by mapping the data into a higher-dimensional feature space.

Example. Find a polynomial of degree $\leq d = 2$ that separates the points $f(z) = \sum \theta_{k_1 \dots k_n} z_1^{k_1} \dots z_n^{k_n}$. For $n = 2$:

$$f(z) = \theta_0 + \theta_1 z_1 + \theta_2 z_2 + \theta_3 z_1^2 + \theta_4 z_2^2 + \theta_5 z_1 z_2$$

The constraints become linear inequalities in θ : $f(x_i) \geq 1$, $f(y_j) \leq -1$. We can solve this using LP or QP to find the coefficients θ .

The Kernel Trick for Non-linear SVMs.

SVM find a linear separating hyperplane. However, many datasets are not linearly separable in the original input space $\mathbb{R}^n$. To handle this, we map the input vectors x into a higher-dimensional *feature space* $\mathcal{H}$ using a mapping function $\Phi : \mathbb{R}^n \rightarrow \mathcal{H}$, $x \mapsto \Phi(x)$. We then find a linear separator in $\mathcal{H}$, which corresponds to a non-linear separator in the original space.

The Dual Problem and Dot Products.

The "Kernel Trick" relies on the observation that the *dual formulation* of the SVM optimization problem depends on the data points only through their *inner products* (dot products).

Dual SVM Problem:

$$\begin{aligned} & \text{maximize} && \sum_{i=1}^N \alpha_i - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \alpha_i \alpha_j y_i y_j (x_i^T x_j) \\ & \text{subject to} && \sum_{i=1}^N \alpha_i y_i = 0 \\ & && 0 \leq \alpha_i \leq C \end{aligned}$$

When mapping to feature space, the term $x_i^T x_j$ is replaced by $\Phi(x_i)^T \Phi(x_j)$.

The Kernel Function.

Computing $\Phi(x)$ directly can be computationally expensive. If we have a *Kernel function* $K(x_i, x_j)$:

$$K(x_i, x_j) = \langle \Phi(x_i), \Phi(x_j) \rangle = \Phi(x_i)^T \Phi(x_j)$$

we can compute the *inner product* in the *high-dimensional* space *without* ever explicitly *computing the coordinates* $\Phi(x)$. This is the **kernel trick**.

Common Kernels:

- *Linear Kernel:* $K(x, z) = x^T z$ (Standard SVM).
- *Polynomial Kernel:* $K(x, z) = (1 + x^T z)^d$.
- *Radial Basis Function (RBF) / Gaussian Kernel:*

$$K(x, z) = \exp(-\gamma \|x - z\|^2)$$

This corresponds to an infinite-dimensional feature space.

Theorem (Mercer). A function $K(x, z)$ is a valid kernel (i.e., corresponds to some dot product in some space) if and only if the Gram matrix $G_{ij} = K(x_i, x_j)$ is positive semidefinite for any set of vectors.

Unconstrained Minimization.

We want to solve the unconstrained convex optimization problem:

$$\text{minimize } f(x)$$

where $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex and differentiable twice continuously. We assume the problem is solvable ($p^* > -\infty$) and the optimal set X^* is nonempty.

General Descent Method.

Algorithm produces a sequence $x^{(k)}$:

$$x^{(k+1)} = x^{(k)} + t^{(k)} \Delta x^{(k)}, \quad \text{where:}$$

- $\Delta x^{(k)}$ is the *search direction*. It must be a descent direction: $\nabla f(x^{(k)})^T \Delta x^{(k)} < 0$.
- $t^{(k)} > 0$ is the *step size* (or step length).

Line Search (Backtracking).

Exact line search (minimizing f along the ray) is usually too expensive. We use *Backtracking*. Parameters: $\alpha \in (0, \frac{1}{2})$, $\beta \in (0, 1)$.

1. Start with $t = 1$.
2. While $f(x + t\Delta x) > f(x) + \alpha t \nabla f(x)^T \Delta x$:
3. $t := \beta t$

This ensures that sufficient decrease is met.

Gradient Descent.

Definition (Gradient descent). Gradient descent *uses the negative gradient as the search direction*:

$$\Delta x = -\nabla f(x)$$

This direction points in the direction of steepest local descent.

Convergence Analysis.

We analyze convergence under the assumption of *strong convexity*. $\exists m, M > 0$ constants such that:

$$mI \preceq \nabla^2 f(x) \preceq MI$$

- M is an upper bound on curvature (Lipschitz continuity of gradient).
- m is a lower bound on curvature.

The ratio $\kappa = M/m$ is the **condition number** of the sublevel sets.

Convergence Rate:

For exact line search (and similar for backtracking), the error contracts linearly:

$$f(x^{(k)}) - p^* \leq c^k (f(x^{(0)}) - p^*)$$

where the constant c depends on the condition number:

$$c = 1 - \frac{m}{M}$$

Unconstrained Minimization: Newton's Method.

Newton's method was originally developed to find roots of functions ($F(x) = 0$). For optimization, we look for a root of the gradient ($\nabla f(x) = 0$).

Definition (Newton step). The Newton step Δx_{nt} at a point x is defined as:

$$\Delta x_{nt} = -\nabla^2 f(x)^{-1} \nabla f(x)$$

where $\nabla^2 f(x)$ (the Hessian) is assumed to be positive definite.

Descent Direction Check:

$$\nabla f(x)^T \Delta x_{nt} = -\nabla f(x)^T \nabla^2 f(x)^{-1} \nabla f(x) < 0$$

unless $\nabla f(x) = 0$. This confirms it is a descent direction.

Interpretations of the Newton Step.

1. **Minimizer of Second-order Approximation:** The second-order Taylor approximation of f at x is:

$$\hat{f}(x+v) = f(x) + \nabla f(x)^T v + \frac{1}{2} v^T \nabla^2 f(x) v$$

Minimizing this quadratic function w.r.t. v :

$$\nabla_v \hat{f}(x+v) = \nabla f(x) + \nabla^2 f(x) v = 0 \implies v = -\nabla^2 f(x)^{-1} \nabla f(x)$$

So Δx_{nt} is the step to the minimum of the local quadratic approximation.

2. **Steepest Descent in Hessian Norm:** Consider the norm defined by the Hessian at x : $\|v\|_{\nabla^2 f(x)} = (v^T \nabla^2 f(x) v)^{1/2}$. The steepest descent direction in this norm is:

$$\Delta x_{nsd} = \operatorname{argmin}\{\nabla f(x)^T v \mid \|v\|_{\nabla^2 f(x)} = 1\}$$

It can be shown that Δx_{nt} points in the same direction as Δx_{nsd} . Specifically, Δx_{nt} is the unnormalized steepest descent step.

3. **Solution to Linearized Optimality Condition:** Optimality condition: $\nabla f(x^*) = 0$. Linearize the gradient near x :

$$\nabla f(x+v) \approx \nabla f(x) + \nabla^2 f(x) v = 0$$

Solving for v gives the Newton step.

Halting Condition: Newton Decrement.

Definition (Newton decrement). We define the Newton decrement $\lambda(x)$ as:

$$\lambda(x) = (\nabla f(x)^T \nabla^2 f(x)^{-1} \nabla f(x))^{1/2}$$

Properties:

- It relates to the quantity minimized by the approximation: $f(x) - \inf_v \hat{f}(x+v) = \frac{1}{2} \lambda(x)^2$.
- It is the length of the Newton step in the Hessian norm: $\lambda(x) = \|\Delta x_{nt}\|_{\nabla^2 f(x)}$.
- Used for stopping criterion: stop if $\lambda(x)^2/2 \leq \epsilon$.

Newton's Algorithm.

1. Given starting point $x \in \text{dom } f$, tolerance $\epsilon > 0$.
2. Repeat:
 - (a) Compute Newton step Δx_{nt} and decrement λ .
 - (b) Stopping criterion: if $\lambda^2/2 \leq \epsilon$, quit.
 - (c) Line search: Choose step size t by backtracking.
 - (d) Update: $x := x + t\Delta x_{nt}$.

Lemma (Affine Invariance). *Newton's method is affine invariant. If we perform a change of coordinates $x = Ty$ (where T is nonsingular), the Newton steps in y -coordinates correspond exactly to the Newton steps in x -coordinates transformed by T^{-1} .*

Proof. Gradient transforms as $\nabla_y \bar{f}(y) = T^T \nabla_x f(x)$. Hessian transforms as $\nabla_y^2 \bar{f}(y) = T^T \nabla_x^2 f(x) T$. The step transforms:

$$\Delta y_{nt} = -(T^T H T)^{-1} (T^T g) = -T^{-1} H^{-1} T^{-T} T^T g = -T^{-1} H^{-1} g = T^{-1} \Delta x_{nt}.$$

Thus $x_{new} = T y_{new}$. □

Convergence Analysis.

Assumptions:

- (I) $mI \preceq \nabla^2 f(x) \preceq MI$ (Strong convexity, bounded curvature).
- (II) $\nabla^2 f(x)$ is Lipschitz continuous with constant L : $\|\nabla^2 f(x) - \nabla^2 f(y)\|_2 \leq L\|x - y\|_2$.

Theorem (Convergence stages). *If gradient is large ($\|\nabla f(x)\|_2 \geq \eta$), the algorithm behaves like gradient descent with constant decrease in f .*

Once the gradient is small enough ($\|\nabla f(x)\|_2 < \eta$), backtracking selects $t = 1$ and convergence becomes quadratic:

$$\frac{\|\nabla f(x^{(k+1)})\|_2}{2m} \leq \left(\frac{\|\nabla f(x^{(k)})\|_2}{2m} \right)^2 \frac{L}{2m^2}$$

The number of iterations to high precision is extremely small (typically 5-6 once in this phase).

Self-Concordance.

The standard convergence analysis depends on constants (m, M, L) which change under affine transformations, contradicting the affine invariance of the method. We want a coordinate-independent analysis.

Definition (Self-concordant). *A convex function $f : \mathbb{R} \rightarrow \mathbb{R}$ is self-concordant if:*

$$|f'''(x)| \leq 2f''(x)^{3/2}$$

For multivariate functions, this must hold for the function restricted to any line.

Example. *Linear and quadratic functions (3rd derivative is 0).*

Log-barrier: $f(x) = -\log x$ is self-concordant.

Properties:

- Self-concordance is preserved under affine transformation and summation.
- *Bound:* If f is self-concordant, the complexity of Newton's method (number of steps) depends only on the parameters of the problem (like number of constraints) and not on condition numbers or unknown constants L, m .

Equality Constrained Minimization.

We consider the convex optimization problem with linear equality constraints:

$$\begin{array}{ll} \text{minimize} & f(x) \\ \text{subject to} & Ax = b \end{array}$$

where $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex and twice continuously differentiable, and $A \in \mathbb{R}^{p \times n}$ with **rank** $A = p < n$. We assume an optimal solution x^* exists and use a feasible starting point $x^{(0)}$ (i.e., $Ax^{(0)} = b$).

Theorem (Optimality Conditions (KKT)). *A point $x^* \in \text{dom } f$ is optimal if and only if there exists a dual variable $\nu^* \in \mathbb{R}^p$ such that:*

1. Primal Feasibility: $Ax^* = b$
2. Stationarity: $\nabla f(x^*) + A^T \nu^* = 0$

The Newton Step.

The Newton step Δx_{nt} is the direction that minimizes the 2nd-order approximation of f at x subject to the equality constraints.

Derivation via Quadratic Approximation:

$$\begin{array}{ll} \text{minimize} & \hat{f}(x+v) = f(x) + \nabla f(x)^T v + \frac{1}{2} v^T \nabla^2 f(x) v \\ \text{subject to} & A(x+v) = b \implies Av = 0 \quad (\text{since } Ax = b) \end{array}$$

The Newton step Δx_{nt} is the solution v to this problem.

Derivation via Linearized Optimality Conditions: We replace the optimality conditions for $x+v$ with their linearization:

$$\begin{aligned} \nabla f(x+v) + A^T w &\approx \nabla f(x) + \nabla^2 f(x)v + A^T w = 0 \\ A(x+v) &= b \implies Av = 0 \end{aligned}$$

This leads to the KKT System:

$$\begin{bmatrix} \nabla^2 f(x) & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} \Delta x_{nt} \\ w \end{bmatrix} = \begin{bmatrix} -\nabla f(x) \\ 0 \end{bmatrix}$$

where w is the optimal dual variable for the quadratic approximation (associated with the constraint). The matrix on the left is called the *KKT matrix*.

Newton Decrement.

The Newton decrement $\lambda(x)$ for equality constrained problems is defined as:

$$\lambda(x) = (\Delta x_{nt}^T \nabla^2 f(x) \Delta x_{nt})^{1/2}$$

Alternatively, it is related to the value of the quadratic approximation:

$$f(x) - \inf\{\hat{f}(x+v) \mid Av = 0\} = \frac{1}{2} \lambda(x)^2$$

Stopping criterion: quit if $\lambda(x)^2/2 \leq \epsilon$.

Algorithm (Newton's Method with Equality Constraints).

Given a starting point $x \in \mathbf{dom} f$ with $Ax = b$, and tolerance $\epsilon > 0$.

1. Compute the Newton step Δx_{nt} and decrement $\lambda(x)$ by solving the KKT system.
2. Stopping criterion: if $\lambda^2/2 \leq \epsilon$, quit.
3. Line search: Choose step size t by backtracking line search.
4. Update: $x := x + t\Delta x_{nt}$.

The algorithm maintains feasibility ($Ax = b$) because $A\Delta x_{nt} = 0$, so $A(x+t\Delta x_{nt}) = Ax+tA\Delta x_{nt} = b$.

Infeasible Start Newton Method.

If we do not have a feasible starting point, we can use Newton's method to simultaneously minimize the objective and reduce the residual $r = Ax - b$. The KKT system is modified to handle the non-zero residual:

$$\begin{bmatrix} \nabla^2 f(x) & A^T \\ A & 0 \end{bmatrix} \begin{bmatrix} \Delta x_{nt} \\ w \end{bmatrix} = \begin{bmatrix} -\nabla f(x) \\ -(Ax - b) \end{bmatrix}$$

This step Δx_{nt} tries to zero out the residual (linearized constraint) while descending the objective.

Interior-Point Methods: The Barrier Method.

We consider the convex optimization problem:

$$\begin{aligned} & \text{minimize} && f_0(x) \\ & \text{subject to} && f_i(x) \leq 0, \quad i = 1, \dots, m \\ & && Ax = b \end{aligned}$$

We assume the problem is strictly feasible (Slater's condition holds), so $\exists x : f_i(x) < 0$.

Logarithmic Barrier Function.

We replace the inequality constraints with a "barrier" in the objective function. The most common choice is the *logarithmic-barrier*:

$$\phi(x) = - \sum_{i=1}^m \log(-f_i(x))$$

The domain of ϕ is the strictly feasible set. As x approaches the boundary (where $f_i(x) \rightarrow 0$), $\phi(x) \rightarrow \infty$.

Approximation of Indicator Function.

The barrier method approximates the problem:

$$\begin{aligned} & \text{minimize} && f_0(x) + \sum_{i=1}^m I_-(f_i(x)) \\ & \text{subject to} && Ax = b \end{aligned}$$

where $I_-(u) = \begin{cases} 0 & u \leq 0, \\ \infty & \text{otherwise.} \end{cases}$ The log barrier $-\frac{1}{t} \sum \log(-f_i(x))$ approximates I_- as $t \rightarrow \infty$.

Definition (Central path). For a parameter $t > 0$, we define $x^*(t)$ as the solution to:

$$\begin{aligned} & \text{minimize} && t f_0(x) + \phi(x) \\ & \text{subject to} && Ax = b \end{aligned}$$

The set of points $\{x^*(t) \mid t > 0\}$ is called the central path.

Optimality Conditions (Centrality Condition).

$x^*(t)$ is strictly feasible and satisfies:

$$t \nabla f_0(x^*(t)) + \nabla \phi(x^*(t)) + A^T \nu = 0$$

Substituting $\nabla \phi(x) = \sum \frac{1}{-f_i(x)} \nabla f_i(x)$:

$$t \nabla f_0(x) + \sum_{i=1}^m \frac{1}{-f_i(x)} \nabla f_i(x) + A^T \nu = 0$$

This looks like the KKT condition for the original problem if we define dual variables $\lambda_i^*(t) = \frac{1}{-t f_i(x^*(t))}$.

Note that $\lambda_i^*(t) f_i(x^*(t)) = -1/t$.

Duality Gap on Central Path: Using these $\lambda^*(t)$, the duality gap is exactly m/t . This means

$$f_0(x^*(t)) - p^* \leq m/t \quad \text{as} \quad t \rightarrow \infty,$$

$x^*(t)$ converges to the optimal solution x^* .

Algorithm (The Barrier Method Algorithm.)

Given strictly feasible x , $t := t^{(0)} > 0$, $\mu > 1$, tolerance $\epsilon > 0$.

1. Compute $x^*(t)$ by minimizing $tf_0(x) + \phi(x)$ subject to $Ax = b$, starting at x . *// centering¹*
2. $x := x^*(t)$ *// update*
3. If $m/t < \epsilon$, quit *// stopping criterion*
4. $t := \mu t$ *// increase t*

Convergence analysis.

The convergence depends on two parts:

1. **Outer Iterations (Updates of t):** The duality gap decreases by a factor of μ at each step. The number of outer iterations to reach precision ϵ is exactly:

$$\frac{\log(m/(\epsilon t^{(0)}))}{\log \mu}$$

2. **Inner Iterations (Newton Steps):** Thanks to self-concordance properties (if functions are self-concordant), Newton's method converges extremely fast (quadratically) to the next point on the central path because $x^*(t)$ is a good starting point for $x^*(\mu t)$.

Total Complexity: The total number of Newton steps grows as $\sqrt{m}$ (theoretically) or is practically constant/logarithmic.

¹Usually using Newton's method

Feasibility and Phase I Methods.

The standard barrier method requires a strictly feasible starting point $x^{(0)}$. If such a point is not known, we must solve a preliminary *Phase I* problem to find one by introducing a slack variable s :

$$\begin{aligned} & \text{minimize} && s \\ & \text{subject to} && f_i(x) \leq s, \quad i = 1, \dots, m \\ & && Ax = b \end{aligned}$$

- **Initialization:** Choose any $x^{(0)}$ satisfying $Ax^{(0)} = b$. Choose $s^{(0)} > \max_i f_i(x^{(0)})$. The pair $(x^{(0)}, s^{(0)})$ is strictly feasible for the Phase I problem.
- **Outcome:** Solve using the barrier method. Let p^* be the optimal value.
 1. If $p^* < 0$: We found a strictly feasible point (since $f_i(x) \leq s < 0$). We can stop early once $s < 0$.
 2. If $p^* > 0$: The original problem is infeasible.
 3. If $p^* = 0$: The feasible set has empty interior (or we converged to the boundary).

Problems with Generalized Inequalities.

We consider the problem with generalized inequality constraints:

$$\begin{aligned} & \text{minimize} && f_0(x) \\ & \text{subject to} && f_i(x) \preceq_{K_i} 0, \quad i = 1, \dots, m \\ & && Ax = b \end{aligned}$$

where $\preceq_{K_i}$ denotes inequality with respect to a proper cone $K_i \subseteq \mathbb{R}^{k_i}$.

Definition (Generalized logarithm). *We need a barrier function for the cone K . A function $\psi : \text{int } K \rightarrow \mathbb{R}$ is a generalized logarithm for K of degree θ if:*

1. ψ is concave and closed.
2. $\psi(ty) = \psi(y) + \theta \log t$ for all $y \succ_K 0$ and $t > 0$.

Example. • Non-negative Orthant ($\mathbb{R}_+^n$): $\psi(x) = \sum \log x_i$. Degree $\theta = n$.

- Second-Order Cone: $\psi(y, t) = \log(t^2 - \|y\|_2^2)$. Degree $\theta = 2$.
- Positive Semidefinite Cone ($\mathbf{S}_+^n$): $\psi(X) = \log \det X$. Degree $\theta = n$.

Barrier Method for Cone Programs.

We define the logarithmic barrier for the problem constraints $f_i(x) \preceq_{K_i} 0$:

$$\phi(x) = - \sum_{i=1}^m \psi_i(-f_i(x))$$

where ψ_i is the generalized logarithm for K_i with degree θ_i .

The Central Path.

Minimizing $tf_0(x) + \phi(x)$ yields points $x^*(t)$ on the central path. The optimality conditions involve dual variables for the conic constraints. The ****Duality Gap**** on the central path is:

$$\frac{\sum_{i=1}^m \theta_i}{t}$$

The convergence analysis is identical to the scalar case, replacing the number of constraints m with the sum of degrees $\bar{\theta} = \sum \theta_i$.